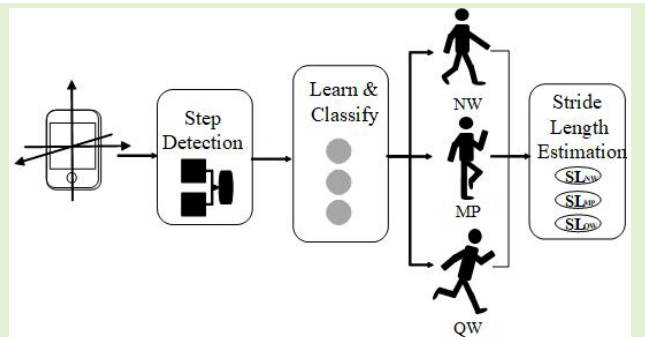


A Robust Step Detection and Stride Length Estimation for Pedestrian Dead Reckoning Using a Smartphone

Yingbiao Yao¹, Lei Pan, Wei Feng¹, Xiaorong Xu¹, *Member, IEEE*, Xuesong Liang, and Xin Xu

Abstract—As an infrastructure-free positioning and navigation method, pedestrian dead reckoning (PDR) is still a research hotspot in the field of indoor localization. Step detection (SD) and stride length estimation (SLE) are two key components of PDR, and it is a challenging problem to apply SD and SLE to different walking patterns. Focusing on this problem, this paper proposes a robust SD and SLE method based on recognizing three walking patterns (i.e., Normal Walk, March in Place, and Quick Walk) using a smartphone. First, we propose a dynamic time warping–based peak prediction with zero-crossing detection to improve the SD accuracy. In particular, the proposed SD can accurately identify the starting and ending points of each step in the three walking patterns. Second, according to the extracted features of each step, a random forest algorithm with classification proofreading is used to recognize the three walking patterns. Finally, an improved SLE model is proposed for the different walking patterns to achieve a higher SLE accuracy. The experimental results show that, on average, the SD accuracy is about 97.9%, the recognition accuracy is about 98.4%, and the relative error of the estimated walking distance is about 3.0%, which outperforms those of the existing commonly used SD and SLE methods.

Index Terms—Pedestrian dead reckoning, step detection, stride length estimation, walking pattern.



I. INTRODUCTION

ALTHOUGH global positioning system (GPS)–based [1] outdoor localization is very common, indoor localization remains a challenge because GPS signals are unavailable indoors. Many indoor localization methods are based on wireless radio systems, such as Wi-Fi [2], radio-frequency identity [3], ultra-wideband [4], and Bluetooth [5]. These localization methods can be categorized into two types [6]: triangulation and fingerprinting. The former relies on installed expensive hardware, making it neither scalable nor universal. The latter requires pretraining, which is time-consuming.

In addition to wireless radio-based methods, pedestrian dead reckoning (PDR) [7], [8], relying on inertial sensors, provides another way for indoor localization. PDR-based methods are advantageous, as they derive the current location by adding the estimated displacement to the previous location, independent

of the external infrastructure. PDR localization errors mainly depend on the accuracy of the step detection (SD), stride length estimation (SLE), and heading direction estimation.

Nowadays, it is still a challenging problem to adapt SD and SLE to different walking patterns. For example, when the pedestrian is in the state of Marching in Place (MP), the traditional SD method has difficulty counting the steps because there are multiple close peaks or valleys in each step. Moreover, when the pedestrian is in the state of MP, Normal Walking (NW), or Quick Walking (QW), the SLE model should be different because these modes have different walking characteristics (e.g., the stride length of the MP pattern should be zero).

The purpose of this paper is to design a robust SD and SLE method that can adapt to the changes in walking patterns, phone poses, and users. Specifically, we consider two phone poses in which the phone is held in front of the chest (Pose1) or put in a trouser pocket (Pose2) and three walking patterns: NW, MP, and QW. NW is the most common walking pattern, and its speed range is usually 1–1.5 m/s. In MP, the pedestrian's two feet move up and down while remaining in place, and it does not cause walking distance. In QW, the pedestrian walks or jogs at a brisk pace, and its speed range is usually 2–2.5 m/s.

Based on the above considerations, we propose a robust SD and SLE method for PDR using a smartphone's built-in

Manuscript received March 27, 2020; accepted April 19, 2020. Date of publication April 23, 2020; date of current version August 5, 2020. This work was supported in part by the Zhejiang Province Science Foundation for Public Welfare under Grant LGG19F020014 and in part by the China Postdoctoral Science Foundation under Grant 2017M621796. The associate editor coordinating the review of this article and approving it for publication was Prof. Sheng-Shian Li. (Corresponding author: Lei Pan.)

The authors are with the School of Communication Engineering, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: 1101351539@qq.com).

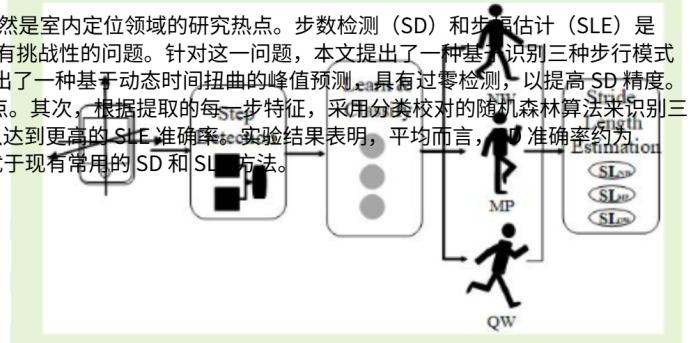
Digital Object Identifier 10.1109/JSEN.2020.2989865

使用智能手机进行行人航位推算的鲁棒步数检测 and 步幅估计

姚颖彪, 潘磊, 冯伟

, 徐晓荣, IEEE 会员, 梁雪松, 徐昕

摘要 — 行人航位推算 (PDR) 作为一种无基础设施的定位和导航方法, 仍然是室内定位领域的研究热点。步数检测 (SD) 和步幅估计 (SLE) 是 PDR 的两个关键组成部分, 将 SD 和 SLE 应用于不同的步行模式是一个具有挑战性的问题。针对这一问题, 本文提出了一种基于识别三种步行模式 (即正常步行、原地行进和快速步行) 的鲁棒 SD 和 SLE 方法。首先, 提出了一种基于动态时间扭曲的峰值预测, 具有过零检测, 以提高 SD 精度。特别是, 所提出的 SD 可以准确识别三种步行模式中每个步骤的起点和终点。其次, 根据提取的每一步特征, 采用分类校准的随机森林算法来识别三种步行模式; 最后, 针对不同的步行模式, 提出了一种改进的 SLE 模型, 以达到更高的 SLE 准确率。实验结果表明, 平均而言, SD 准确率约为 97.9%, 识别准确率约为 98.4%, 估计步行距离的相对误差约为 3.0%, 优于现有常用的 SD 和 SLE 方法。



索引术语 — 行人航位推算、步数检测、步幅估计、步行模式。

一、我

尽管基于全球定位系统 (GPS) [1] 的室外定位非常普遍, 但室内定位仍然是一个挑战, 因为 GPS 信号在室内不可用。许多室内定位方法基于无线电系统, 如 Wi-Fi [2]、射频识别 [3]、超宽带 [4] 和蓝牙 [5]。这些定位方法可分为两种类型 [6]: 三角测量和指纹识别。前者依赖于安装昂贵的硬件, 使其既不可扩展, 也不可通用。后者需要预训练, 这非常耗时。除了基于无线电的方法外, 行人航位推算 (PDR) [7]、[8] 依赖惯性传感器, 为室内定位提供了另一种方法。PDR-b 定位方法具有优势, 因为它们通过将估计的位移与先前的位置相加来推导当前位置, 这是独立的

外部基础设施的定位。PDR 定位误差主要取决于步数检测 (SD)、步幅估计 (SLE) 和航向估计的准确性。

如今, 使 SD 和 SLE 适应不同的步行模式仍然是一个具有挑战性的问题。例如, 当行人处于原地行进 (MP) 状态时, 传统的 SD 方法很难计算步数, 因为每一步都有多个接近的峰值或低谷。此外, 当行人处于 MP、正常步行 (NW) 或快速步行 (QW) 状态时, SLE 模型应该是不同的, 因为这些模式具有不同的步行特征 (例如, MP 模式的步幅应为零)。

本文的目的是设计一种能够适应行走模式、手机姿势和用户变化的鲁棒 SD 和 SLE 方法。具体来说, 我们考虑了两种手机姿势, 即手机放在胸前 (Pose1) 或放入裤袋 (Pose2) 和三种步行模式: NW、MP 和 QW。NW 是最常见的步行模式, 其速度范围通常为 1–1.5 m/s。在 MP 中, 行人的两只脚在保持原位的同时上下移动, 并且不会造成步行距离。在 QW 中, 行人以较快的速度行走或慢跑, 其速度范围通常为 2–2.5 m/s。

基于上述考虑, 我们提出了一种使用智能手机内置的 PDR 的鲁棒 SD 和 SLE 方法

稿件于 2020 年 3 月 27 日收到; 2020 年 4 月 19 日接受。发表日期: 2020 年 4 月 23 日; 当前版本发布日期: 2020 年 8 月 5 日。这项工作得到了浙江省公益科学基金 LGG19F020014 资助和中国博士后科学基金资助 2017M621796 资助。协调本文审稿并批准发表的副主编是李胜世安教授。 通讯作者: 潘磊。

作者来自杭州 310018 杭州电子工业大学通信工程学院 (电子邮件: 1101351539@qq.com)。

数字对象标识符 10.1109/JSEN.2020.2989865

1558-1748 © 2020 年 IEEE。允许个人使用, 但重新出版/重新分发需要 IEEE 许可。

有关详细信息, 请参阅 <https://www.ieee.org/publications/rights/index.html>。

授权许可使用仅限于: 华中科技大学。于 2025 年 9 月 17 日 03: 52: 14 UTC 从 IEEE Xplore 下载。有限制。

inertial sensors. The key idea behind the proposed method is to recognize the pedestrian's walking patterns and then to select the SLE model according to the recognized walking patterns, which can improve the existing SD and SLE methods' robustness and practicability. To our knowledge, the contributions of this paper can be summarized as follows:

- We propose a robust SD method, which combines a dynamic time warping (DTW)-based peak prediction with zero-crossing detection, to accurately detect the step time boundaries in different walking patterns.
- We propose a random forest (RF)-based walking pattern-aware SLE method to improve the SLE accuracy which, using new features and classification proofreading (CP), can recognize the walking patterns of each step, and then use our improved SLE models for different walking patterns.
- We compare our method with the existing SD and SLE methods and demonstrate that our method outperforms the commonly used methods.

The rest of the paper is organized as follows. Section II introduces the related work. The components and design details of our proposed method are described in Section III. Then, we present the experiment results and analysis in Section IV. Finally, Section V concludes this paper.

II. RELATED WORK

A. Step Detection

Generally, the existing SD algorithms fall into 3 categories: time-domain methods [9]–[14], frequency-domain methods [15]–[17], and feature-clustering methods [18]–[20]. Since our method belongs to the time-domain methods, we focus on the SD methods in the time domain. The time-domain methods include threshold detection [9], peak detection [10], zero-crossing detection [11], correlation detection [12], etc. Threshold detection counts steps by judging whether sensor data surpasses predefined thresholds. However, though the thresholding method is the simplest, it is difficult to preset an optimal threshold for all cases. Peak detection estimates steps based on the number of peaks in a sequence of sensor data, but it suffers from interference peaks due to noise and occasional disturbance. Zero-crossing detection measures steps by detecting the number of zero points in sensor data, but there are many false zero points due to interference and noise. Nowadays, due to the complexity of actual sensor signals, it is difficult to achieve satisfactory SD accuracy with only one method. Therefore, combination methods are proposed to improve the SD accuracy. In [13], Zhang *et al.* combined zero-crossing and peak detection to limit the interval between two zero points and reduce false zero points. In [14], the DTW method [21] is adopted for evaluating the correlation of three-axis accelerations, and it can limit the problems resulting from naturally occurring changes in walking speed.

B. Walking Pattern Recognition

Using inertial sensors to sense human activity, including pedestrian walking patterns, has been explored in academia.

Susi *et al.* [22] proposed a decision tree-based classifier to recognize four motion modes. WTrack proposed a hidden Markov model-based walking pattern recognition [23] with a total of 72 patterns, in which walking patterns are the combination of walking speed and direction. In [24], with a support vector machine and decision tree, combinations of motion states and phone poses can be recognized successfully. Catal *et al.* [25] utilized the J48 decision tree, multilayer perceptron, and logistic regression techniques and combined these classifiers with the average of probabilities combination rule to recognize human activities. Miyamoto and Ogawa [26] used histograms of oriented gradients to extract features and used Real AdaBoost as a classifier. In [27], the authors demonstrated the feasibility of using the newly emerging barometer in modern smartphones to distinguish vertical walking patterns, such as upstairs and downstairs. Recently, deep learning-based activity recognition has been getting a lot of attention. Hassan *et al.* [28] proposed a deep belief network-based approach for human activity recognition. Gu *et al.* [29] adopted a stacked denoising autoencoder-based deep learning method for locomotion activity recognition using data from multiple sensors available on most smart devices. In [30], the authors employed a deep learning method to recognize different walking speeds and phone-carrying methods using stacked autoencoders. Zhou *et al.* [31] proposed smartphone-based activity recognition using a new convolutional neural network, which can identify nine activities for use as landmarks for indoor localization.

C. Stride Length Estimation

Assuming stride length is proportional to the vertical displacement of the hip at each step, Weinberg proposed a famous SLE model in [32], which is based on the maximum and minimum acceleration within each step. Alvarez *et al.* [33] modeled stride length as a function of height and leg length by considering human gait as an inverted pendulum model. Wei *et al.* [34] proposed an SLE system based on computer vision to assess the movement of Parkinson's patients. In [35], an SLE method based on the wavelet transform is presented. Shin *et al.* [36] proposed an adaptive SLE method according to the pedestrian's motion status, that is, walking or running. Double integration is another important approach for SLE algorithms [37]. However, due to intrinsic measurement errors in the currently available inertial sensors, minimizing the integration drift of low-precision sensors in smartphones remains an open question. With the recent advances in machine learning, several machine learning-based SLE approaches have been proposed. For example, a deep convolutional neural network and an online sequential extreme learning machine were used to estimate stride length in [38], [39] and [13], respectively.

III. THE PROPOSED METHOD

The architecture of our method, shown in Figure 1, consists of four modules: data preprocessing, SD, walking pattern recognition (WPR), and SLE. The contributions of this paper cover three modules: SD, WPR, and SLE. In the SD module,

惯性传感器。所提方法的关键思想是识别行人的行走模式, 然后根据识别的行走模式选择 SLE 模型, 这可以提高现有 SD 和 SLE 方法的鲁棒性和实用性。据我们所知, 本文的贡献可以总结如下:

- 我们提出了一种鲁棒 SD 方法, 该方法将基于动态时间扭曲 (DTW) 的峰值预测与过零检测相结合, 以准确检测不同行走模式下的步长时间边界。
- 我们提出了一种基于随机森林 (RF) 的步行模式感知 SLE 方法来提高 SLE 的准确性, 该方法利用新功能和分类校对 (CP), 可以识别每一步的步行模式, 然后使用我们改进的 SLE 模型来处理不同的步行模式。
- 我们将我们的方法与现有的 SD 和 SLE 方法进行了比较, 并证明我们的方法优于常用方法。

本文的其余部分组织如下。第二节介绍了相关工作。我们所提出的方法的组成部分和设计细节在第三节中进行了描述。然后, 我们在第四节中介绍了实验结果和分析。最后, 第五节结束了本文。

二、RW。

A. 步数检测

通常, 现有的 SD 算法分为 3 类: 时域方法[9]–[14]、频域方法[15]–[17]和特征聚类方法[18]–[20]。由于我们的方法属于时域方法, 因此我们重点关注时域中的 SD 方法。时域方法包括阈值检测[9]、峰值检测[10]、过零检测[11]、相关检测[12]等。阈值检测通过判断传感器数据是否超过预定义的阈值来计算步长。然而, 尽管阈值方法最简单, 但很难为所有情况预设一个最佳阈值。峰值检测根据传感器数据序列中的峰值数来估计步长, 但由于噪声和偶尔的干扰, 它会受到干扰峰值的影响。过零检测通过检测传感器数据中的零点数量来测量步长, 但由于干扰和噪声, 假零点较多。如今, 由于实际传感器信号的复杂性, 仅靠一种方法很难达到令人满意的标清精度。因此, 提出了组合方法来提高标清精度。在[13]中, Zhang 等将过零和峰值检测相结合, 以限制两个零点之间的间隔并减少假零点。在[14]中, 采用 DTW 方法[21]来评估三轴加速度的相关性, 它可以限制步行速度自然变化引起的问题。

B. 步行模式识别

学术界已经探索了使用惯性传感器来感知人类活动, 包括行人行走模式。

Susi 等[22]提出了一种基于决策树的分类器来识别四种运动模式。WTrack 提出了一种基于隐马尔可夫模型的步行模式识别[23], 共有 72 种模式, 其中步行模式是步行速度和方向的组合。在[24]中, 通过支持向量机和决策树, 可以成功识别运动状态和手机姿势的组合。Catal 等[25]利用 J48 决策树、多层感知器和逻辑回归技术, 并将这些分类器与概率组合规则的平均值相结合, 以识别人类活动。Miyamoto 和 Ogawa [26]使用定向梯度的直方图来提取特征, 并使用 Real AdaBoost 作为分类器。在[27]中, 作者证明了在现代智能手机中使用新出现的气压计来区分垂直行走模式 (如楼上和楼下) 的可行性。最近, 基于深度学习的活动识别引起了广泛关注。Hassan 等[28]提出了一种基于深度信念网络的人类活动识别方法。Gu 等[29]采用了一种基于堆叠去噪自编码器的深度学习方法, 利用大多数智能设备上可用的多个传感器的数据进行运动活动识别。在[30]中, 作者采用了一种深度学习方法来识别不同的步行速度, 并使用堆叠的自动编码器来识别手机携带方法。周等[31]提出了使用一种新的卷积神经网络进行基于智能手机的活动识别, 该网络可以识别九种活动, 用作室内定位的地标。

C. 步幅估计

假设步幅与每一步髋关节的垂直位移成正比, Weinberg 在[32]中提出了一个著名的 SLE 模型, 该模型基于每步内的最大和最小加速度。Alvarez 等[33]通过将人体步态视为倒摆模型, 将步幅建模为身高和腿长的函数。Wei 等[34]提出了一种基于计算机视觉的 SLE 系统来评估帕金森病患者的运动。在[35]中, 提出了一种基于小波变换的 SLE 方法。Shin 等[36]提出了一种根据行人的运动状态 (即步行或跑步) 的自适应 SLE 方法。双积分是 SLE 算法的另一种重要方法[37]。然而, 由于目前可用的惯性传感器存在固有测量误差, 最小化智能手机中低精度传感器的积分漂移仍然是一个悬而未决的问题。随着机器学习的最新进展, 人们提出了几种基于机器学习的 SLE 方法。例如, 在[38]、[39]和[13]中, 深度卷积神经网络和在线顺序极限学习机分别用于估计步幅。

三、TPM

我们的方法的架构如图 1 所示, 由四个模块组成: 数据预处理、SD、行走模式识别 (WPR) 和 SLE。本文的贡献涵盖三个模块: SD、WPR 和 SLE。在 SD 模块中,

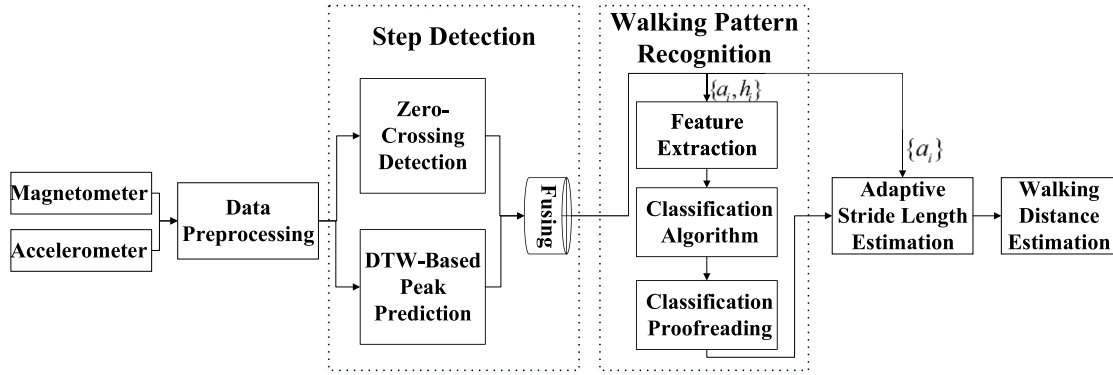


Fig. 1. The architecture of our method.

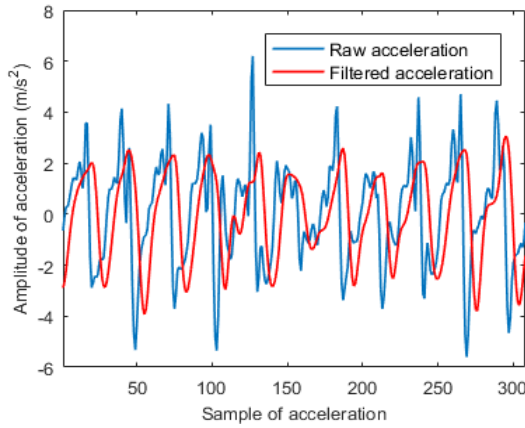


Fig. 2. The performance of the 4th-order Butterworth filter.

a DTW-based peak prediction is combined with zero-crossing detection to accurately detect the step boundaries. In the WPR module, the RF algorithm with the new features and CP is used to recognize the walking patterns. In the SLE module, two state-of-the-art SLE methods are fused.

A. Data Preprocessing

Data preprocessing is used to ease noise and interference effects. The original derived accelerometer data includes X, Y, and Z values, and we use the acceleration amplitude to detect the step event, as follows:

$$a_t = \sqrt{a_{x,t}^2 + a_{y,t}^2 + a_{z,t}^2} - g \quad (1)$$

where $a_{x,t}$, $a_{y,t}$, and $a_{z,t}$ are the accelerometer readings of the X-axis, Y-axis, and Z-axis acquired at time t , respectively, and g is the gravity component, equal to 9.8 m/s^2 .

The raw acceleration may contain interference and noise signals, which can deteriorate the subsequent SD and WPR performance. Therefore, a 4th-order Butterworth digital low-pass filter, with a cutoff frequency of $w_c = 0.2\pi$, is applied to filter the high frequency component of a_t , and its filtering effect is shown in Figure 2. It should be pointed out that although there is a delay in the filtered signal in Figure 2, this delay has no other effect except that the output results lag.

B. Step Detection

The goals of the SD include: 1) it can meet the requirements of step counting under different walking patterns, and 2) it can correctly segment the data of each step, laying a foundation for the subsequent walking pattern recognition. To achieve these two goals, the proposed SD method combines DTW-based peak prediction with cross-zero detection. The former realizes accurate step counting under different walking patterns, and the latter can identify the starting point of each step.

1) **Zero-Crossing Detection:** Park *et al.* [40] showed that arm and foot movements are synchronized during walking. Therefore, the starting or ending point of each step should be a zero point in the acceleration signal, according to the gait model in [41]. However, due to the influence of the sampling rate, walking speed, and noise, the sampled acceleration signal may have no zero point, or the zero point may not be the true starting or ending point. Hence, we recorded the point passing through the zero as the zero point. All potential starting and ending points, determined by zero-crossing detection, are marked as set $S1$, as shown in (2):

$$S1 = \{t | a_t \times a_{t-1} \leq 0\} \quad (2)$$

However, the zero-crossing detection method generally has a problem with false zero points. Therefore, we combine zero-crossing detection with peak detection to exclude false zero points from the set $S1$.

2) **DTW-Based Peak Prediction:** Peak detection is a common method for SD. However, it also suffers from false peaks, especially when a pedestrian is in the MP pattern. An example is shown in Figure 3, in which the number of steps using the traditional peak detection method will be much greater than the real number, due to false peaks. To solve this problem, the DTW-based peak prediction method is proposed to calculate the similarity of two adjacent steps' acceleration signals and then predict the range of the true peak.

The DTW-based peak prediction method consists of three key modules: the periodicity constraint, DTW constraint, and searching maximum peak, as shown in Figure 4.

a) **Periodicity constraint:** In general, the periods of two adjacent steps in the same walking pattern have similarities. Here, the period refers to the difference between two

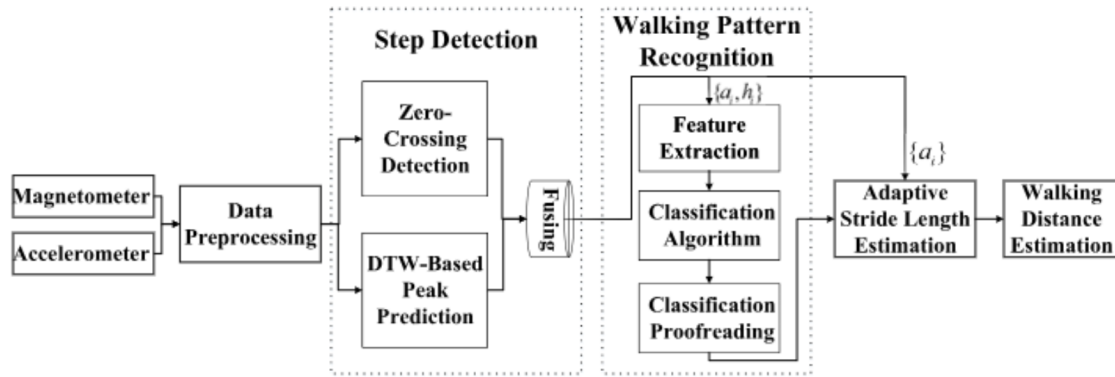


图 1. 我们方法的架构。

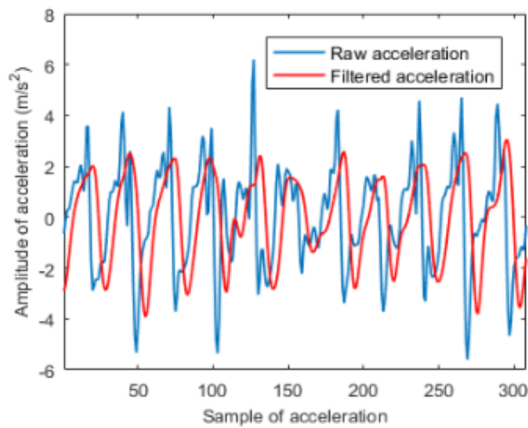


图 2.4 阶巴特沃斯滤波器的性能。

基于 DTW 的峰值预测与过零检测相结合，准确检测步进边界。在 WPR 模块中，使用具有新功能和 CP 的 RF 算法来识别行走模式。在 SLE 模块中，融合了两种最先进的 SLE 方法。

A. 数据预处理

数据预处理用于缓解噪声和干扰影响。原始导出的加速度计数据包括 X、Y 和 Z 值，我们使用加速度幅度来检测步进事件，如下所示：

$$\sqrt{a_x^2 + a_y^2 + a_z^2 - g^2} \quad (1)$$

其中 a_x 、 a_y 和 a_z 分别是在时间 t 获取的 X 轴、Y 轴和 Z 轴的加速度计读数， g 是重力分量，等于 9.8 m/s。

原始加速度可能包含干扰和噪声信号，这会降低后续的 SD 和 WPR 性能。因此，应用截止频率为 $w=0.2\pi$ 的 4 阶巴特沃斯数字低通滤波器对 a 的高频分量进行滤波，其滤波效果如图 2 所示。需要指出的是，虽然图 2 中滤波后的信号存在延迟，但这种延迟除了输出结果滞后外没有其他影响。

B. 步数检测

SD 的目标包括：1) 能够满足不同步行模式下的步数计数要求，2) 能够正确分割每一步的数据，为后续的步行模式识别奠定基础。为了实现这两个目标，所提出的 SD 方法将基于 DTW 的峰值预测与过零检测相结合。前者实现了不同步行模式下的精确步数计数，后者可以识别每一步的起点。

1) 过零检测：Park 等[40]表明，在行走过程中，手臂和脚的运动是同步的。因此，根据[41]中的步态模型，每一步的起点或终点应为加速度信号中的零点。然而，由于采样率、行走速度和噪声的影响，采样的加速度信号可能没有零点，或者零点可能不是真正的起点或终点。因此，我们将穿过零点的点记录为零点。通过过零检测确定的所有潜在起点和终点都标记为集合 $S1$ ，如 (2) 所示：

$$S1 = \{t | a \times a \leq 0\} \quad (2)$$

然而，过零检测方法普遍存在假零点的问题。因此，我们将过零检测与峰值检测相结合，从集合 $S1$ 中排除假零点。

2) 基于 DTW 的峰值预测：峰值检测是一种常见的 SD 方法。然而，它也存在假峰值，特别是当行人处于 MP 模式时。图 3 所示为一个示例，其中由于假峰值，使用传统峰值检测方法的步数将远大于实数。为了解决这个问题，提出了基于 DTW 的峰值预测方法，计算两个相邻步骤的加速度信号的相似度，然后预测真峰值的范围。

基于 DTW 的峰值预测方法由三个关键模块组成：周期性约束、DTW 约束和搜索最大峰值，如图 4 所示。

a) 周期性约束：一般来说，同一步行模式中两个相邻步骤的周期具有相似性。这里，周期是指两个之间的差异

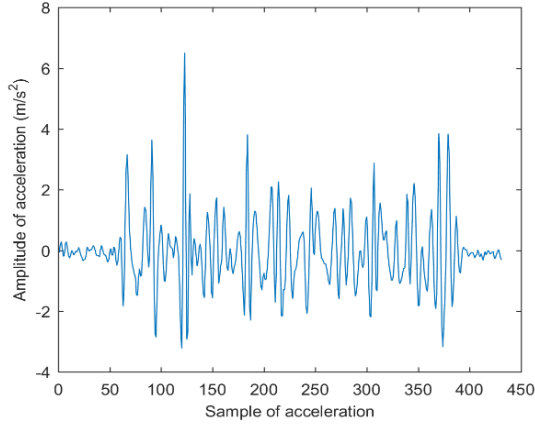


Fig. 3. Acceleration waveform of 10 steps in the MP pattern.

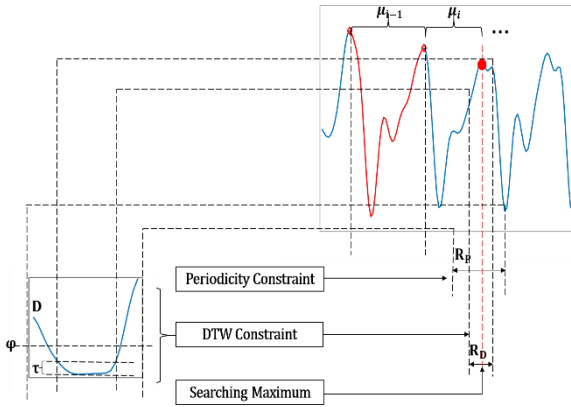


Fig. 4. The DTW-based peak prediction.

consecutive true peak moments, which is defined as follows:

$$t_i = t_{i+1}^{peak} - t_i^{peak} \quad (3)$$

where t_i^{peak} is the moment of the i -th peak, and t_i is the period of the i -th step.

The periods' similarity indicates that each period in the same walking pattern is very close to the next. Therefore, the next step's period should satisfy the following periodic similarity constraint:

$$t_i \in ((1 - \theta) * T_{i-1}, (1 + \theta) * T_{i-1}) \quad (4)$$

where θ is the similarity parameter, and T_i is the period template. To improve the accuracy of T_i , it is updated as follows:

$$T_i = \frac{T_{i-1} + t_i}{2} \quad (5)$$

According to the periodic constraints, we can initially derive the range of the next peak, $\mathbf{R}_p$, as shown in Figure 4.

b) *DTW constraint*: DTW was proposed to calculate the distance or similarity of two sequences with different lengths in the time domain [21]. In this paper, we employed DTW to predict the time range of the current step's true peak acceleration.

Algorithm 1 DTW-Based Peak Prediction Method

Input: The acceleration $\{a\}$, the latest two peak moments t_{i-1}^{peak} , t_i^{peak} , the period template T_{i-1} .

Output: The next peak moment t_{i+1}^{peak} , the next period template T_i .

1. $\mathbf{R}_p = \emptyset$; /* Initialize $\mathbf{R}_p$ */
2. **FOR** each moment j
3. **IF** $j - t_i^{peak} \in ((1 - \theta) * T_{i-1}, (1 + \theta) * T_{i-1})$
4. $\mathbf{R}_p = \mathbf{R}_p \cup \{j\}$
5. **ENDIF**
6. **ENDFOR**
7. $\mu_{i-1} = a(t_{i-1}^{peak} : t_i^{peak})$; /* Extract the matched sequence */
8. **FOR** each moment k of $\mathbf{R}_p$
9. $v_k = a(t_i^{peak} : k)$; /* Build the matching sequence */
10. $\mathbf{D}(k) = DTW(\mu_{i-1}, v_k)$
11. **ENDFOR**
12. $\mathbf{R}_D = \emptyset$; /* Initialize $\mathbf{R}_D$ */
13. **FOR** each moment k of $\mathbf{R}_p$
14. **IF** $\mathbf{D}(k) < \varphi$ & $\mathbf{D}(k) - \min(\mathbf{D}) < \tau$
15. $\mathbf{R}_D = \mathbf{R}_D \cup \{k\}$
16. **ENDIF**
17. **ENDFOR**
18. $t^{peak} = \arg \max_{t \in \mathbf{R}_D} (a(t))$
19. $t_i = t_{i+1}^{peak} - t_i^{peak}$
20. $T_i = \frac{T_{i-1} + t_i}{2}$

We consider the acceleration sequence between the latest two peaks as the matched sequence μ , as shown in (6). In particular, the first matched sequence μ_0 is obtained using the traditional peak detection method.

$$\mu_{i-1} = a(t_{i-1}^{peak} : t_i^{peak}) \quad (6)$$

We take the acceleration data from the latest peak to the first sample point in $\mathbf{R}_p$ as the first test sequence v_1 , the acceleration data from the latest peak to the second sample point in $\mathbf{R}_p$ as the second test sequence v_2 , and so on, to obtain a test sequence set $\mathbf{V} = \{v_1, v_2, \dots, v_j, \dots, v_k\}$. Then, the DTW distance of μ and v_j , $DTW(\mu, v_j)$, is calculated to obtain a DTW distance set $\mathbf{D} = \{DTW(\mu, v_1), DTW(\mu, v_2), \dots, DTW(\mu, v_k)\}$. According to the similarity of each step in the same walking pattern, the DTW distance will be significantly small when the latest sample point in the test sequence is near the next true peak. Hence, the sample point j in $\mathbf{R}_p$, which satisfies the condition in (7), is outputted to the candidate true peak set, $\mathbf{R}_D$, which is the prediction range of the next true peak.

$$\mathbf{R}_D = \{j | j \in \mathbf{R}_p, DTW(\mu, v_j) < \varphi,$$

$$DTW(\mu, v_j) - D_{\min} < \tau\} \text{ and } D_{\min} = \min(\mathbf{D}) \quad (7)$$

where φ and τ are used to control the number of elements in $\mathbf{R}_D$. In addition, since the DTW constraint is to utilize the

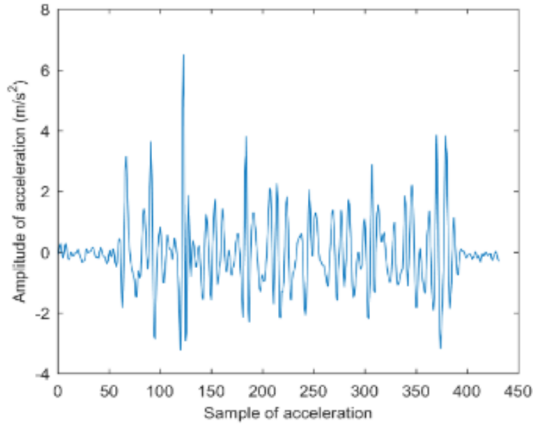


图 3.MP 模式中 10 步的加速波形。

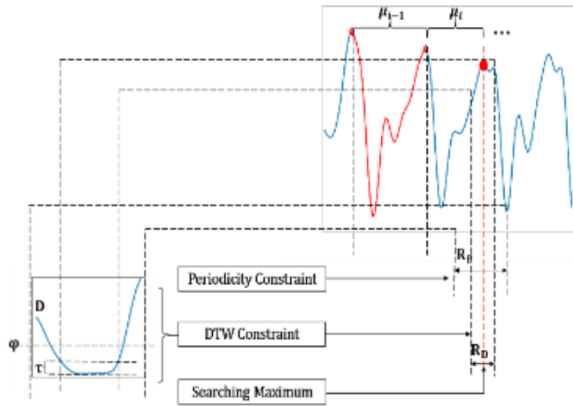


图 4.基于 DTW 的峰值预测。

连续的真实峰值矩, 定义如下:

$$t = t_{i+1} - t_i \quad (3)$$

基第 t 个峰值的时刻, 是第 i 个步骤的周期。

周期的相似性表示同一行走模式中的每个周期都与下一个周期非常接近。因此, 下一步的周期应满足以下周期相似性约束:

$$t \in ((1-\theta) * T, (1+\theta) * T) \quad (4)$$

其中 θ 是相似性参数, T 是周期模板。为了提高 T 的准确性, 它更新如下:

$$T = \frac{T + t}{2} \quad (5)$$

根据周期性约束, 我们可以初步推导出下一个峰值 R 的范围, 如图 4 所示。

b) DTW 约束: DTW 用于计算两个不同长度序列在时域中的距离或相似度[21]。在本文中, 我们采用 DTW 来预测当前步骤的真实峰值加速度的时间范围。

算法 1 基于 DTW 的峰值预测方法输入: 加速度 $\{a\}$, 最近两个峰值矩 t

我-1, i 吨, 期间模板 T 。

输出: 下一个峰值时刻 t_{i+1} 、下一个期间模板 T 。

1. $R = \emptyset$; /* 初始化 R */
2. 为了每一刻 j
3. 如果 $j - t_i \in ((1-\theta) * T, (1+\theta) * T)$
4. $R = R \cup \{j\}$
5. 恩迪夫
6. END 为的
7. $\mu = \text{一个 } (t : t_i)$; /* 提取匹配的序列 */
8. 对于 R 的每一刻 k
9. $v = \text{一个 } t_i : k$; /* 构建匹配序列 */
10. $D(k) = \text{DTW}(\mu, v)$
11. END 为的
12. $R = \emptyset$; /* 初始化 R */
13. 对于 R 的每一刻 k
14. 如果 $D(k) < \phi$ 且 $D(k) - \text{最小值}(D) < \tau$
15. $R = R \cup \{k\}$
16. 恩迪夫
17. END 为的
18. $t = \text{最大参数}_{t \in R}(a(t))$
19. $t = t_i - t_i$
20. $T = \frac{T + t}{2}$

我们将最新两个峰之间的加速序列视为匹配序列 μ , 如 (6) 所示。特别是, 使用传统的峰检测方法获得第一个匹配序列 μ 。

$$\mu = a(t_i : t_{i+1}) \quad (6)$$

我们取 R_{as} 中从最新峰值到第一个采样点的加速度数据为第一个测试序列 v , 从 R_{as} 中最新峰值到第二个采样点的加速度数据为第二个测试序列 v , 依此类推, 得到测试序列集 $V = \{v_1, v_2, \dots, v_n\}$ 。然后, 计算 μ 和 v 的 DTW 距离, $\text{DTW}(\mu, v)$, 得到 DTW 距离集 $D = \{\text{DTW}(\mu, v_1), \text{DTW}(\mu, v_2), \dots, \text{DTW}(\mu, v_n)\}$ 。根据同一行走模式中每一步的相似性, 当测试序列中的最新采样点接近下一个真峰时, DTW 距离将明显变小。因此, 将 R 中满足 (7) 中条件的采样点 j 输出到候选真峰集 R , 即下一个真峰的预测范围。

$$R = \{j | j \in R, \text{DTW}(\mu, v) < \phi, \text{DTW}(\mu, v) - D_{\min} < \tau \text{ 和 } D = \text{最小值}(D)\} \quad (7)$$

其中 ϕ 和 τ 用于控制 R 中的元素数量。此外, 由于 DTW 约束是利用

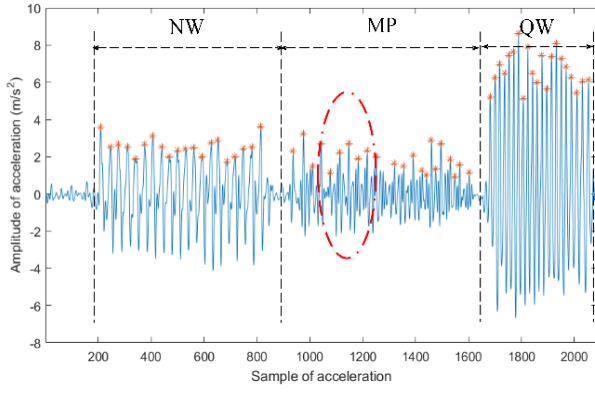


Fig. 5. Results of the DTW-based peak prediction.

TABLE I
PROCESS OF DTW-BASED PEAK PREDICTION

Moment of current peak	Prediction range of next peak
1043	(1071,1083)
1077	(1106,1120)
1113	(1143,1152)
1147	(1170,1186)
1183	(1207,1223)
1217	(1244,1260)

similarity of each step in the same walking pattern, the $\mathbf{R}_D$ will be empty when the pedestrian changes the walking pattern.

c) *Searching Maximum Peak*: After obtaining the next true peak's prediction range, $\mathbf{R}_D$, the maximum acceleration in $\mathbf{R}_D$ corresponds to the true peak moment, as shown in (8).

$$t^{peak} = \arg \max_{t \in \mathbf{R}_D} (a(t)) \quad (8)$$

where t^{peak} represents the current true peak moment.

Algorithm 1 shows the process of the DTW-based peak prediction in detail, and its improvements are: 1) the parameters θ , φ , and τ are the same for the NW, MP, and QW patterns; 2) T_i and μ are automatically established and updated according to the similarity of each step in the same walking pattern; 3) once a pattern change is detected, μ_0 will be reestablished. Therefore, our DTW-based peak prediction method can adapt to different walking patterns.

An example is given in Figure 5 for detecting the true peak in three walking patterns. From Figure 5, we find that the DTW-based peak prediction method accurately detects the peak in NW, MP, and QW.

Furthermore, Table I shows the detailed peak prediction at the red circle marked in Figure 5. When the current peak moment is 1043, we obtain the prediction range of the next peak (1071, 1083), from which 1077 is searched as the moment of the next peak. Then we take 1077 as the current peak moment to predict the range of the next peak, and so on. This process will be repeated until all peaks are accurately detected.

3) *Fusing*: In this paper, the peak moment is obtained using the DTW-based peak prediction method, while the valley moment is the moment corresponding to the minimum acceleration value between two peak moments. Then, the peak

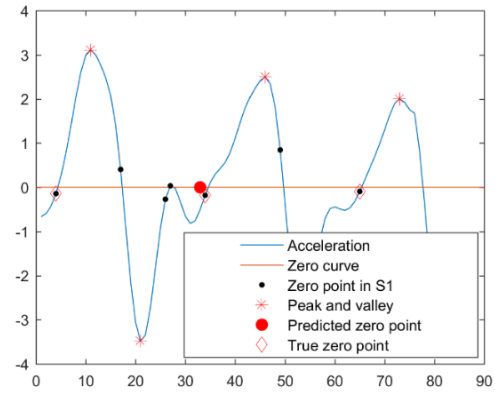


Fig. 6. Example of eliminating the false zeros.

and valley moments are used to predict the true zero-point moment, t_{zero}^i , which is defined in (9):

$$t_{zero}^i = \frac{(t_{i-1}^{valley} + t_i^{peak})}{2} \quad (9)$$

where t_i^{peak} and t_{i-1}^{valley} are the current peak and last valley moments, respectively.

As depicted in Figure 6, we obtain the true zero-point set, $\mathbf{S}_2$, through finding the point in $\mathbf{S}_1$ that is the closest to the predicted zero point, where $\mathbf{S}_1$ is the set of zero points determined by zero-crossing detection, as shown in Equ. (2). Assuming that $\mathbf{S}_1 = \{t(1), t(2), t(3), \dots, t(i), \dots\}$ and t_{zero} is a predicted zero point, then t is a true zero point:

$$\mathbf{S}_2 = \left\{ t | t = \arg \min_{t(i) \in \mathbf{S}_1} (|t(i) - t_{zero}|) \right\} \quad (10)$$

Then the starting and ending points of each step can be expressed as $\{t_i, t_{i+1}\}$, where $t_i, t_{i+1} \in \mathbf{S}_2$, $i = 1, 2, 3, \dots$. From each pair of starting and ending points, we can extract each step's complete information, which forms the basis of feature analysis and feature extraction for WPR.

C. Walking Pattern Recognition

1) *Feature Extraction*: Based on the step detection above, the data of each step from the starting point to the ending point can be obtained. Then, the following features, shown in Table II, are extracted. Moreover, apart from the acceleration data, each step's magnetometer data is also used for feature extraction to increase the recognition accuracy. In Table II, a and h are the amplitude of the acceleration and magnetometer data, respectively; p is the amplitude of the acceleration data power spectrum; $\mathbf{E}[\cdot]$ is for expectation; $\mathbf{I}[\cdot]$ indicates function; and $\mathbf{I}[\cdot] = 1$ when the condition is true; otherwise it is 0.

Some features in Table II are from [28], [42]–[45], and some features (i.e., F1, F8, F9, F12–F15, and F22–F29) are newly proposed in this paper. Here, we only introduce four newly proposed features that pass the subsequent feature selection: F1 (μ_a), F13 (FC), F12 ($diff_{median}$), and F8 (ρ_{zero}). These four features are all extracted from the acceleration signal. Specifically, μ_a is the mean of the absolute value, FC is the center frequency of the power spectrum amplitude, $diff_{median}$ is the difference between the peak and median, and

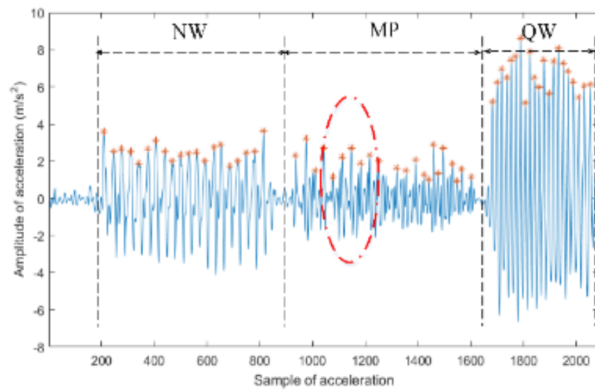


图 5.基于 DTW 的峰值预测结果。

表一

P DTW-BPP

Moment of current peak	Prediction range of next peak
1043	(1071,1083)
1077	(1106,1120)
1113	(1143,1152)
1147	(1170,1186)
1183	(1207,1223)
1217	(1244,1260)

同步行模式中每个步骤的相似性，当行人改变步行模式时，R 将为空。

c) 搜索最大峰值：在获得下一个真实峰值的预测范围 R 后，R 中的最大加速度对应于真实峰值矩，如 (8) 所示。

$$t = \underset{t \in R}{\text{最大参数}} (a(t)) \quad (8)$$

其中 t 代表当前真实峰值时刻。

算法 1 详细展示了基于 DTW 的峰值预测过程，其改进之处在于：1) NW、MP 和 QW 模式的参数 θ 、 ϕ 和 τ 相同；2) T 和 μ 根据同一行走模式中每个步骤的相似性自动建立和更新；3) 一旦检测到模式变化， μ 将被重新建立。因此，我们基于 DTW 的峰值预测方法可以适应不同的行走模式。

图 5 给出了一个示例，用于检测三种行走模式中的真实峰值。从图 5 中，我们发现基于 DTW 的峰值预测方法可以准确检测 NW、MP 和 QW 中的峰值。

此外，表 I 显示了图 5 中标记的红色圆圈处的详细峰值预测。当当前峰值时刻为 1043 时，我们得到下一个峰值的预测范围

(1071、1083)，从中搜索 1077 作为下一个峰值的时刻。然后我们以 1077 作为当前峰值时刻来预测下一个峰值的范围，依此类推。将重复此过程，直到准确检测到所有峰值。

3) 熔断：本文采用基于 DTW 的峰值预测方法获得峰值矩，而谷矩是两个峰值矩之间最小加速度值对应的矩。然后，峰值

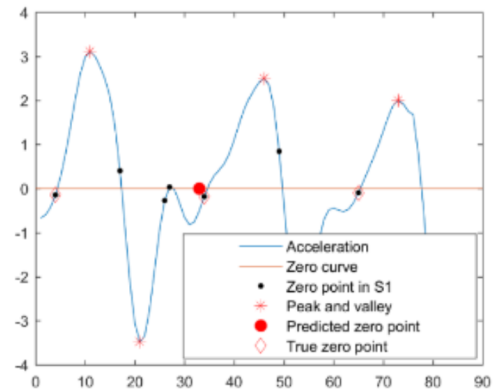


图 6.消除假零的示例。

谷矩用于预测真正的零点矩 t，其定义见 (9)：

$$t_i = \frac{(t_{i-1} + t_{i+1})}{2} \quad (9)$$

其中 t_i 和 t_{i+1} 分别是当前峰值和最后谷值时刻。

如图 6 所示，我们通过找到 S1 中最接近预测零点的点来获得真正的零点集 S2，其中 S1 是通过过零检测确定的零点集，如公式所示。假设 $S1 = \{t(1), t(2), t(3), \dots, t(i), \dots\}$ 并且 t 是预测的零点，则 t 是真正的零点：

$$S2 = \{t | t = \underset{t(i) \in S1}{\text{参数最小值}} |t(i) - t|\} \quad (10)$$

那么，每个步骤的起点和终点可以表示为 $\{t_i, t_{i+1}\}$ ，其中 $t_i, t_{i+1} \in S2$ ， $i = 1, 2, 3, \dots$ 。从每对起点和终点中，我们可以提取出每个步骤的完整信息，这构成了 WPR 特征分析和特征提取的基础。

C. 步行模式识别

1) 特征提取：基于上述步骤检测，可以得到从起点到终点的每一步数据。然后，提取以下特征，如表二所示。此外，除了加速度数据外，还使用每个步骤的磁力计数据进行特征提取，以提高识别精度。在表二中，a 和 h 分别是加速度和磁力计数据的幅度；p 是加速度数据功率谱的幅度；E[] 表示期望值；I[] 表示函数；当条件为真时，I[] = 1；否则为 0。

表 II 中的一些特征来自 [28]、[42]–[45]，以及本文新提出的一些特征（即 F1、F8、F9、F12–F15 和 F22–F29）。在这里，我们只介绍了四个新提出的通过后续特征选择的特征：F1 (μ)、F13 (FC)、F12 (diff) 和 F8 (ρ)。这四个特征都是从加速度信号中提取的。具体来说， μ 是绝对值的平均值，FC 是功率谱幅度的中心频率，diff 是峰值和中值之间的差值，并且

TABLE II
THE EXTRACTED FEATURES

Feature index	Definition
F1	The mean of the absolute acceleration value, $\mu_a = \text{mean}(a_i)$
F2	The acceleration variance, $\sigma_a^2 = \text{var}(a_i)$ [28]
F3	The acceleration skewness, $\text{ske}_a = E\left[\left((a_i - \text{mean}(a_i))/\sqrt{\text{var}(a_i)}\right)^3\right]$ [28]
F4	The acceleration kurtosis, $\text{kur}_a = E\left[\left((a_i - \text{mean}(a_i))/\sqrt{\text{var}(a_i)}\right)^4\right]$ [28]
F5	The magnetic field strength variance, $\sigma_h^2 = \text{var}(h_i)$ [41]
F6	The magnetic field strength skewness, $\text{ske}_h = E\left[\left((h_i - \text{mean}(h_i))/\sqrt{\text{var}(h_i)}\right)^3\right]$
F7	The magnetic field strength kurtosis, $\text{kur}_h = E\left[\left((h_i - \text{mean}(h_i))/\sqrt{\text{var}(h_i)}\right)^4\right]$
F8	The number of zero points in each step, $\rho_{\text{zero}} = \sum_{j=1}^L \mathbb{I}[S1(j) \in (t_i, t_{i+1})]$
F9	The number of local maximum points in each step, $\rho_{\text{extreme}} = \sum_{t=t_i}^{t_{i+1}} \mathbb{I}[a_t > a_{t+1} \& \& a_t > a_{t-1}]$
F10	The difference between two successive peaks of acceleration, $\text{diff}_{\text{peak}} = a(t_{i+1}^{\text{peak}}) - a(t_i^{\text{peak}})$
F11	The period duration, $\text{diff}_{\text{time}} = t_{i+1}^{\text{peak}} - t_i^{\text{peak}}$ [43]
F12	The difference of the peak and median in each step, $\text{diff}_{\text{median}} = a(t_i^{\text{peak}}) - \text{median}(a_i)$
F13	The center frequency, $FC = \sum_{i=1}^N f_i p_i / \sum_{i=1}^N p_i$
F14	The mean square frequency, $MSF = \sum_{i=1}^N f_i^2 p_i / \sum_{i=1}^N p_i$
F15	The frequency variance, $VF = \sum_{i=1}^N (f_i - FC)^2 p_i / \sum_{i=1}^N p_i$
F16-F18	The mean of the absolute value of acceleration of the X-axis, Y-axis, and Z-axis [40]
F19-F21	The variance of acceleration of the X-axis, Y-axis, and Z-axis [40]
F22-F24	The skewness of acceleration of the X-axis, Y-axis, and Z-axis
F25-F27	The kurtosis of acceleration of the X-axis, Y-axis, and Z-axis
F28	Correlation coefficient between X-axis and Y-axis acceleration,
F29	Correlation coefficient between X-axis and Z-axis acceleration,
F30	Correlation coefficient between Y-axis and Z-axis acceleration [42]

ρ_{zero} is the number of zero points in each step. F1 and F13 are effective at recognizing the smartphone pose and walking patterns. F12 and F8 are effective at recognizing the walking patterns.

2) Classification Algorithm: Aiming at the classification problem in this paper, we have implemented three classification algorithms: back propagation neural networks, support vector machine and RF. Experiment results show that RF has the advantages of high accuracy and low complexity, so RF is used to recognize the smartphone poses and walking patterns. We trained three RF classifiers in this paper, as shown in Figure 7. Classifier 1 is used to recognize the smartphone pose

(i.e., Pose1: holding in front of the chest, and Pose2: carrying in the trouser pocket, as shown in Figure 8). Classifiers 2 and 3 are used for WPR (i.e., NW, MP, and QW) in Pose1 and Pose2, respectively.

3) Feature Selection: To make a tradeoff between the accuracy and efficiency of the above three classifiers, we use the Relief-F method in [46], [47] to select the suitable features instead of adopting all 30 features in Table II.

The Relief-F results of the three classifiers are shown in Figure 9. We finally selected the following features, shown in Table III, for each classifier.

表二
TE F

Feature index	Definition
F1	The mean of the absolute acceleration value, $\mu_a = \text{mean}(a_i)$
F2	The acceleration variance, $\sigma_a^2 = \text{var}(a_i)$ [28]
F3	The acceleration skewness, $\text{ske}_a = E\left[\left((a_i - \text{mean}(a_i))/\sqrt{\text{var}(a_i)}\right)^3\right]$ [28]
F4	The acceleration kurtosis, $\text{kur}_a = E\left[\left((a_i - \text{mean}(a_i))/\sqrt{\text{var}(a_i)}\right)^4\right]$ [28]
F5	The magnetic field strength variance, $\sigma_h^2 = \text{var}(h_i)$ [41]
F6	The magnetic field strength skewness, $\text{ske}_h = E\left[\left((h_i - \text{mean}(h_i))/\sqrt{\text{var}(h_i)}\right)^3\right]$
F7	The magnetic field strength kurtosis, $\text{kur}_h = E\left[\left((h_i - \text{mean}(h_i))/\sqrt{\text{var}(h_i)}\right)^4\right]$
F8	The number of zero points in each step, $\rho_{\text{zero}} = \sum_{j=1}^L \text{I}[S1(j) \in (t_i, t_{i+1})]$
F9	The number of local maximum points in each step, $\rho_{\text{extreme}} = \sum_{i=t_i}^{t_{i+1}} \text{I}[a_i > a_{i+1} \& \& a_i > a_{i-1}]$
F10	The difference between two successive peaks of acceleration, $\text{diff}_{\text{peak}} = a(t_{i+1}^{\text{peak}}) - a(t_i^{\text{peak}})$
F11	The period duration, $\text{diff}_{\text{time}} = t_{i+1}^{\text{peak}} - t_i^{\text{peak}}$ [43]
F12	The difference of the peak and median in each step, $\text{diff}_{\text{median}} = a(t_i^{\text{peak}}) - \text{median}(a_i)$
F13	The center frequency, $FC = \sum_{i=1}^N f_i p_i / \sum_{i=1}^N p_i$
F14	The mean square frequency, $MSF = \sum_{i=1}^N f_i^2 p_i / \sum_{i=1}^N p_i$
F15	The frequency variance, $VF = \sum_{i=1}^N (f_i - FC)^2 p_i / \sum_{i=1}^N p_i$
F16-F18	The mean of the absolute value of acceleration of the X-axis, Y-axis, and Z-axis [40]
F19-F21	The variance of acceleration of the X-axis, Y-axis, and Z-axis [40]
F22-F24	The skewness of acceleration of the X-axis, Y-axis, and Z-axis
F25-F27	The kurtosis of acceleration of the X-axis, Y-axis, and Z-axis
F28	Correlation coefficient between X-axis and Y-axis acceleration,
F29	Correlation coefficient between X-axis and Z-axis acceleration,
F30	Correlation coefficient between Y-axis and Z-axis acceleration [42]

ρ 是每步中的零点个数。F1 和 F13 可有效识别智能手机姿势和步行模式。F12 和 F8 可有效识别步行模式。

2) 分类算法: 针对分类

在这篇文章中, 我们实现了三种分类算法: 反向传播神经网络、支持向量机和射频。实验结果表明, 射频具有高精度和低复杂度的优点, 因此使用射频来识别智能手机的姿态和行走模式。本文训练了三种射频分类器, 如图 7 所示

分类器 1 用于识别智能手机姿势 (即 Pose1: 抱在胸前, Pose2: 放在裤兜里, 如图 8 所示)。分类器 2 和 3 分别用于 Pose1 和 Pose2 中的 WPR (即 NW、MP 和 QW)。

3) 特征选择: 为了在上述三个分类器的精度和效率之间进行权衡, 我们使用[46]、[47]中的 Relief-F 方法来选择合适的特征, 而不是采用表 II 中的所有 30 个特征。

三个分类器的 Relief-F 结果如图 9 所示。我们最终为每个分类器选择了以下特征, 如表 III 所示。

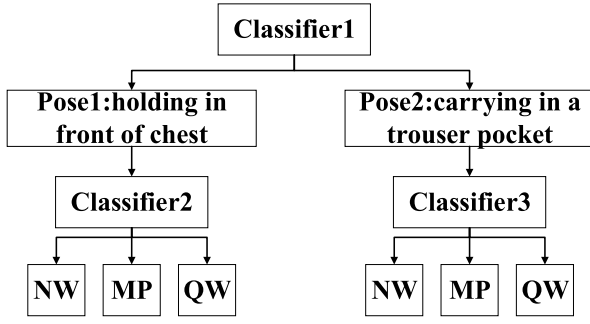


Fig. 7. The structure diagram of classifier 1.

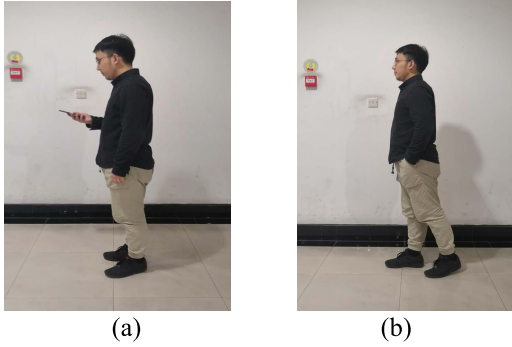


Fig. 8. Two phone poses: (a) holding in front of the chest and (b) carrying in a trouser pocket.

TABLE III
THE SELECTED FEATURE FOR EACH CLASSIFIER

Classifier	Selected features
Classifier1	F17, F18, F16, F13, F1, F20, F10, F14, F21, F23
Classifier2	F1, F13, F16, F19, F12, F8, F14, F2, F9, F11
Classifier3	F1, F20, F11, F2, F4, F12, F28, F29, F30, F16

4) Classification Proofreading: In general, a pedestrian's walking pattern lasts for a period of time instead of changing immediately. Therefore, the adjacent steps' recognition results can be used to improve the walking patterns' recognition accuracy. Specifically, if the recognition result of the current step i is different from that of step $i-1$, the recognition decision is given at most three steps later. We consider the recognition result of the subsequent steps $i+1$, $i+2$, and $i+3$. If they are the same as that of step i , then the recognition result of step i is credible, and we take the classifier result as the final walking pattern. Otherwise, we maintain the walking pattern of step $i-1$ in step i , as shown in (11):

$$WP(i) = \begin{cases} c(i) & \text{if } c(i) = c(i+1) \\ & = c(i+2) = c(i+3) \\ WP(i-1) & \text{otherwise} \end{cases} \quad (11)$$

where $WP(i)$ is the final recognized walking pattern of step i , and $c(i)$ is the classification result of step i . Equ. (11) illustrates that only when four consecutive steps are identified

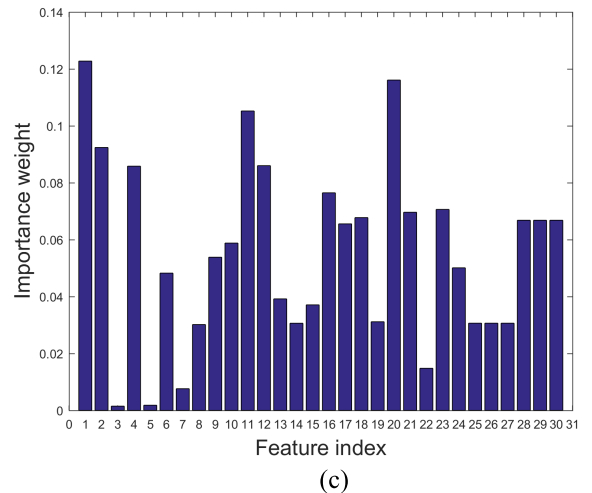
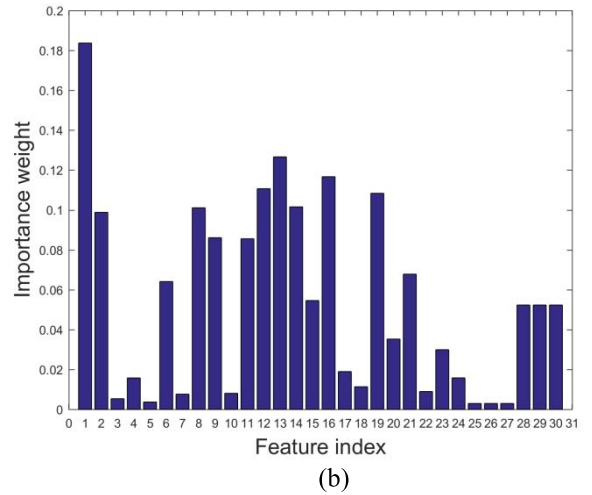
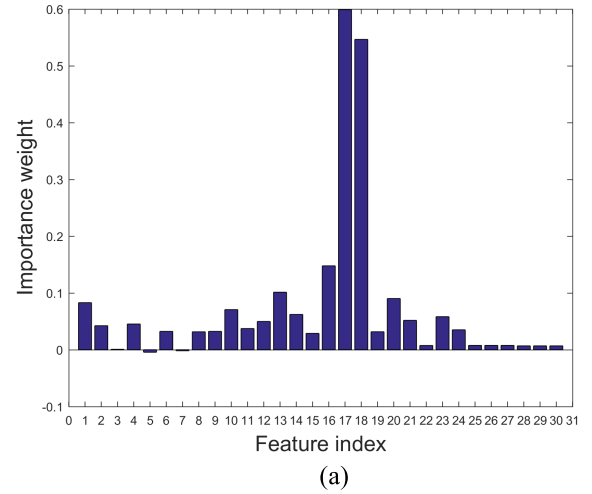


Fig. 9. The Relief-F results of the three classifiers: (a) classifier1, (b) classifier2, and (c) classifier3.

as the same walking pattern does our method consider the walking pattern to have changed.

D. Adaptive Stride Length Estimation

In general, stride length differs between persons and is related to a variety of factors, such as the pedestrian's height, weight, and physical condition. Even for the same person,

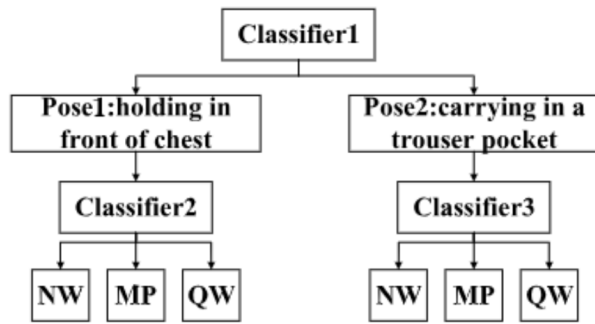


图 7.分类器 1 的结构图。

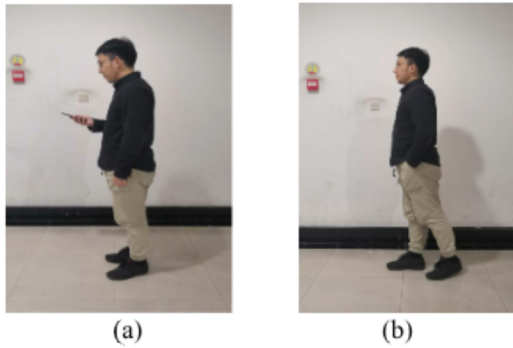


图 8.两种手机姿势： (a) 抱在胸前， (b) 放在裤袋里。

表三

特斯弗埃克

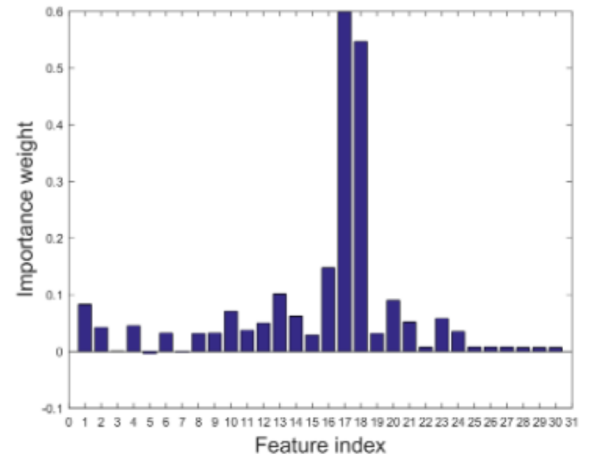
Classifier	Selected features
Classifier1	F17, F18, F16, F13, F1, F20, F10, F14, F21, F23
Classifier2	F1, F13, F16, F19, F12, F8, F14, F2, F9, F11
Classifier3	F1, F20, F11, F2, F4, F12, F28, F29, F30, F16

4) 分类校对：一般来说，行人的步行模式会持续一段时间，而不是立即改变。因此，可以利用相邻步骤的识别结果来提高步行模式的识别准确率。具体来说，如果当前步骤 i 的识别结果与步骤 $i-1$ 的识别结果不同，则最多在三步后给出识别决策。我们考虑后续步骤 $i+1$ 、 $i+2$ 和 $i+3$ 的识别结果。如果它们与步骤 i 相同，那么步骤 i 的识别结果是可信的，我们把分类器结果作为最终的步行模式。否则，我们在步骤 i 中保持步骤 $i-1$ 的步行模式，如 (11) 所示：

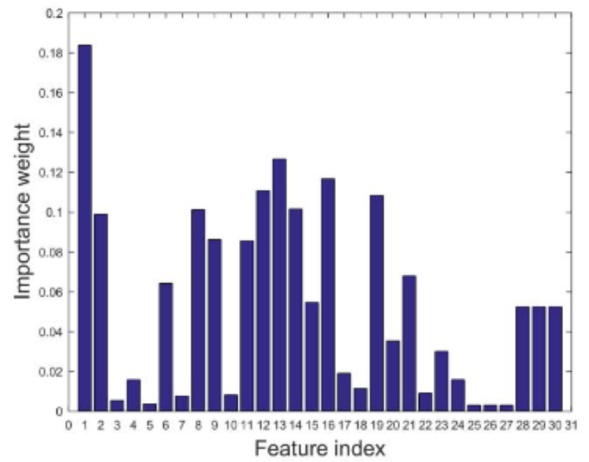
$$WP(i) = \begin{cases} c(i) & \text{我 } f_c(i) = c(i+1) \\ & = c(i+2) = c(i+3) \end{cases} \quad (11)$$

WP(i-1) 否则

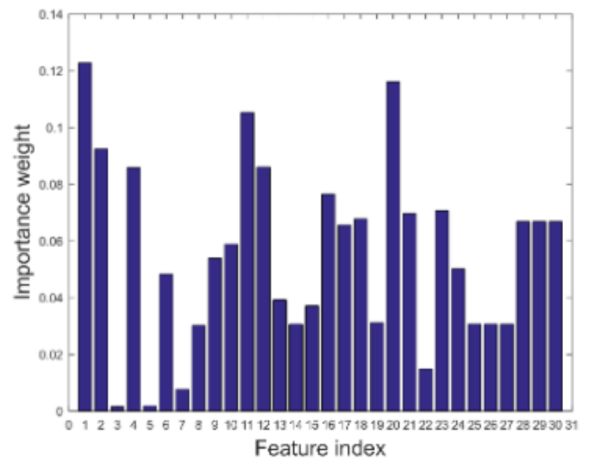
式中， $WP(i)$ 是步骤 i 的最终识别的行走模式， $c(i)$ 是步骤 i 的分类结果。等式 (11) 说明只有当识别出四个连续的步骤时



(a)



(b)



(c)

图 9.三个分类器的 Relief-F 结果： (a) 分类器 1、 (b) 分类器 2 和 (c) 分类器 3。

由于相同的步行模式，我们的方法认为步行模式已经改变。

D. 自适应步幅估计

一般来说，步幅因人而异，与行人的身高、体重、身体状况等多种因素有关。即使对于同一个人，

the stride length of each step in the walking process will be different. In particular, when a pedestrian changes his or her walking pattern, the change in stride length is also obvious. For example, the stride length should be zero when the pedestrian is in the MP pattern. Therefore, we propose a walking pattern-aware SLE method to adapt to three walking patterns: NW, MP, and QW. Moreover, to make the proposed SLE method irrelevant to specific pedestrians, the pedestrians' physiological parameters are not included in our proposed SLE method. Our proposed SLE model combines two popular SLE methods [36], [39], as shown in (12):

$$L = \alpha * \sqrt[4]{a_{\max,k} - a_{\min,k}} + \beta * f_k + \gamma * v_k \quad (12)$$

where α , β , and γ are step parameters that can be determined by the training data, $a_{\max,k}$ and $a_{\min,k}$ represent the maximum and minimum of the acceleration at step k , and f_k and v_k are the pedestrian's walking frequency and the variance of acceleration signals, respectively, which can be calculated as follows:

$$f_k = \frac{1}{t_{k+1} - t_k} \quad (13)$$

$$v_k = \frac{1}{N_k} \sum_{t=t_k}^{t_{k+1}} (a_t - \bar{a}_k)^2 \quad (14)$$

where t_k, t_{k+1} is the starting and ending time of step k , a_t is the acceleration at time t , $\bar{a}_k$ represents the average acceleration of step k , and N_k denotes the number of samples during step k .

The step parameters vary with the walking patterns, and each walking pattern corresponds to a set of step parameters. There is no doubt that the stride length of MP is 0 and, hence, α , β , and γ are all zero in the MP pattern. The step parameters in the NW and QW patterns can be obtained using the least mean square algorithm. When taking SL as the stride length in the real case, then the sum of squared errors of n steps will be:

$$E = \sum_{i=1}^n (SL_i - (\alpha * \sqrt[4]{a_{\max,i} - a_{\min,i}} + \beta * f_i + \gamma * v_i))^2 \quad (15)$$

Taking $X = [x_1, x_2, \dots, x_n]$, where $x_i = [\sqrt[4]{a_{\max,i} - a_{\min,i}}, f_i, v_i]^T$, as the input vector, $D = [SL_1, SL_2, \dots, SL_n]^T$ as the expectation vector, then Equ.(15) can be expressed as:

$$E = (D - X^T W)^T (D - X^T W) \quad (16)$$

where $W = (\alpha, \beta, \gamma)^T$ are the estimated step parameters. At last, according to the least mean square algorithm, W can be calculated as follows:

$$W = [X X^T]^{-1} [X D] \quad (17)$$

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

To evaluate our proposed method, a series of experiments were conducted in the first and sixth buildings of Hangzhou Dianzi University, whose indoor layout is shown in Figure 10.

TABLE IV
USER PROFILE

USER No.	GENDER	HEIGHT (CM)	WEIGHT(KG)	PHONE
1	F	155	49	MATE 8
2	M	168	74	NEXUS 5
3	M	168	60	MATE 8
4	M	178	70	NEXUS 5
5	M	170	66	MATE 8
6	M	172	64	NEXUS 5
7	M	181	69	MATE 8
8	F	162	52	NEXUS 5

The experimental data collection method is similar to that of [29]. Eight volunteers were asked to collect data using two smartphones (Huawei Mate 8 and Google Nexus 5). Table IV shows the volunteers' specific information. Both smartphones have built-in 3-axis accelerometers and 3-axis magnetometers with a sampling frequency of 50 Hz. During the data collection, the volunteers were asked to remember how many steps they took in each walking pattern. The data collection includes the training and testing data collections. In the process of collecting the training data, each volunteer was asked to walk at a constant speed for a total of 100 steps in three walking patterns (MP, NW, and QW) using two phone poses (Pose1 and Pose2) along two paths. We did not strictly measure the volunteers' walking speeds, and we only asked them to perform MP, NW, and QW according to their own habits. Therefore, there are 96 trajectories, each of which corresponds to one pattern, one phone pose, and one path. When a volunteer walks at a constant pace, his/her step length for each step is approximately the same. The ground-truth step length for the SLE can then be obtained by dividing the length of the path by the number of steps walked. In the testing data collection, the volunteers were asked to perform a continuous motion with a fixed number of steps along another path three times. The continuous motion consists of three walking patterns using two phone poses. Therefore, there are 24 trajectories, each including three walking modes and two phone poses. We use a tuple $\langle \text{walking pattern, phone pose, the number of steps} \rangle$ for describing a part of the continuous motion. Therefore, the order of the walking patterns in the testing path is $\langle \text{NW, Pose1, 20} \rangle \rightarrow \langle \text{MP, Pose1, 20} \rangle \rightarrow \langle \text{NW, Pose1, 20} \rangle \rightarrow \langle \text{QW, Pose1, 20} \rangle \rightarrow \langle \text{MP, Pose2, 30} \rangle \rightarrow \langle \text{NW, Pose2, 30} \rangle \rightarrow \langle \text{QW, Pose2, 30} \rangle$.

B. Step Detection Evaluation

We compared the performance of our proposed SD method with Zhang's [13] and Gu's algorithms [48]. The former combines the zero-crossing and peak detection methods to limit the interval between two zero points and reduce the incidence of false zero points. The latter applied periodicity, similarity, and continuity to the peak detection to eliminate false peaks and to improve the step-counting accuracy. In addition, the parameters of our DTW-based SD method are determined according to the training data, as shown in Table V. These parameters are independent of the pedestrian, smartphone, and walking pattern but related to the sensor's sampling frequency.

在步行过程中, 每一步的步幅都会有所不同。特别是, 当行人改变其行走模式时, 步幅的变化也很明显。例如, 当行人处于 MP 模式时, 步幅应为零。因此, 我们提出了一种步行模式感知 SLE 方法, 以适应三种步行模式: NW、MP 和 QW。此外, 为了使所提出的 SLE 方法与特定行人无关, 我们提出的 SLE 方法中不包括行人的生理参数。我们提出的 SLE 模型结合了两种流行的 SLE 方法[36]、[39], 如 (12) 所示:

$$L = \alpha * \sqrt{a - a + \beta * f + \gamma * v} \quad (12)$$

其中 α 、 β 和 γ 是可以通过训练数据确定的步进参数, a 和 a 表示步长 k 处加速度的最大值和最小值, f 和 v 分别是行人的行走频率和加速度信号的方差, 可以计算如下:

$$f = \frac{1}{t - t} \quad (13)$$

$$v = \frac{1}{N} \sum_{t=t}^{} (\text{一个} - \text{一个}) \quad (14)$$

式中, t 为步骤 k 的起止时间, a 为时间 t 时的加速度, a 表示步骤 k 的平均加速度, N 表示步骤 k 期间的样本数。

步参数随步行模式而变化, 每个步行模式对应一组步参数。毫无疑问, MP 的步幅为 0, 因此, α 、 β 和 γ 在 MP 模式中都是零。NW 和 QW 模式中的步进参数可以使用最小均方算法获得。当在实际情况下以 SL 作为步幅时, 则 n 步的平方误差之和为:

$$E = \sum_{i=1}^n (S \cdot L \cdot (\alpha * \sqrt{a - a + \beta * f + \gamma * v})) \quad (15)$$

服用 $\sqrt{X} = [x, x, \dots, x]$, 哪里 $x =$ 一个 - 一个, F, v , 如这输入向量 $D = [SL, SL, \dots, SL]$ 作为期望向量, 则 Equ. (15) 可以表示为:

$$E = (D - X \text{ 宽}) (D - X \text{ 宽}) \quad (16)$$

其中 $W = (\alpha, \beta, \gamma)$ 是估计的步长参数。最后, 根据最小均方算法, W 可以计算如下:

$$W = \frac{[X \text{ 宽}]}{[X \text{ 宽}] [X \text{ 宽}]} \quad (17)$$

IV. ERA. 实验设置

为了评估我们所提出的方法, 在杭州电子科技大学的第一和第六栋楼中进行了一系列实验, 其室内布局如图 10 所示。

表四
UP

USER NO.	GENDER	HEIGHT (CM)	WEIGHT(KG)	PHONE
1	F	155	49	MATE 8
2	M	168	74	NEXUS 5
3	M	168	60	MATE 8
4	M	178	70	NEXUS 5
5	M	170	66	MATE 8
6	M	172	64	NEXUS 5
7	M	181	69	MATE 8
8	F	162	52	NEXUS 5

实验数据收集方法与[29]类似。8 名志愿者被要求使用两部智能手机 (华为 Mate 8 和 Google Nexus 5) 收集数据。表 IV 显示了志愿者的具体信息。这两款智能手机都内置了 3 轴加速度计和 3 轴磁力计, 采样频率为 50 Hz。在数据收集过程中, 志愿者被要求记住他们在每种步行模式中走了多少步。数据收集包括训练和测试数据收集。在收集训练数据的过程中, 每个志愿者被要求沿着两条路径使用两个手机姿势 (Pose1 和 Pose2) 以三种步行模式 (MP、NW 和 QW) 匀速行走共 100 步。我们没有严格测量志愿者的步行速度, 我们只要求他们根据自己的习惯进行 MP、NW 和 QW。因此, 有 96 条轨迹, 每条轨迹对应一种模式、一种手机姿势和一条路径。当志愿者以恒定的速度行走时, 他/她每一步的步长大致相同。然后可以通过将路径长度除以步行的步数来获得 SLE 的地面实况步长。在测试数据收集中, 志愿者被要求沿着另一条路径以固定步数执行连续运动 3 次。连续运动由使用两个手机姿势的三种步行模式组成。因此, 有 24 条轨迹, 每条轨迹包括三种步行模式和两种手机姿势。我们使用一个元组<步行模式、手机姿势、步数>来描述连续运动的一部分。因此, 测试路径中行走模式的顺序为 $\rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow$ 。

B. 步进检测评估

我们将所提出的 SD 方法的性能与 Zhang 的[13]和 Gu 的算法[48]进行了比较。前者结合了过零和峰值检测方法, 以限制两个零点之间的间隔并减少假零点的发生率。后者将周期性、相似性和连续性应用于峰值检测, 以消除假峰值并提高计步精度。此外, 我们基于 DTW 的 SD 方法的参数是根据训练数据确定的, 如表 V 所示。这些参数与行人、智能手机和步行模式无关, 但与传感器的采样频率有关。

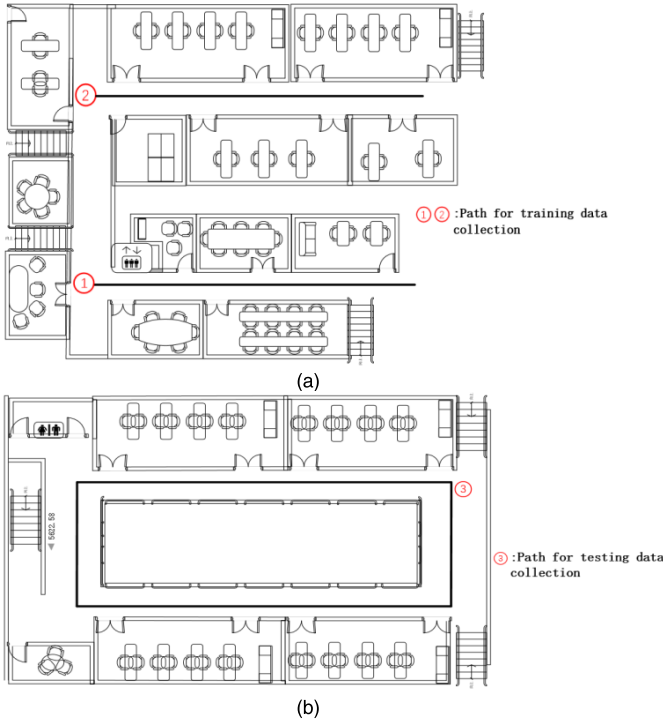


Fig. 10. The indoor layout of the experiment site. (a) The training data collection occurred in the first building. (b) The testing data collection occurred in the sixth building.

TABLE V
PARAMETER SETTING OF OUR DTW-BASED SD METHOD

Parameters	value
φ	40
τ	20
θ	0.4

We adopted the relative error to compare the SD performance, which is defined as follows:

$$SD_e = \frac{|N_e - N_t|}{N_t} \times 100\% \quad (18)$$

where N_e and N_t are the estimated value and the true value, respectively. We calculated the average relative error of all the testing trajectories, and the results are shown in Figure 11.

Figure 11 shows that our SD method is better than the two comparison methods. Zhang's method is obviously not suitable for MP because the MP error rate is significantly higher than that of the other two walking patterns. In addition, the change in smartphone pose during walking invalidated the threshold in Zhang's method, which led to the error in the QW pattern. Although Gu's method has an acceptable detection accuracy in the NW pattern, its SD results worsened in the MP and QW patterns. Therefore, both Zhang's and Gu's methods cannot adapt to different walking patterns. Our proposed method uses DTW for peak prediction, combines periodic constraints and adaptive thresholds to adapt to different walking patterns, and

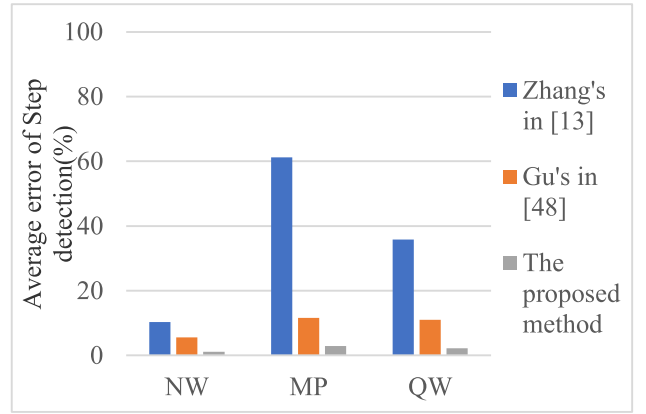


Fig. 11. Step detection performance comparison.

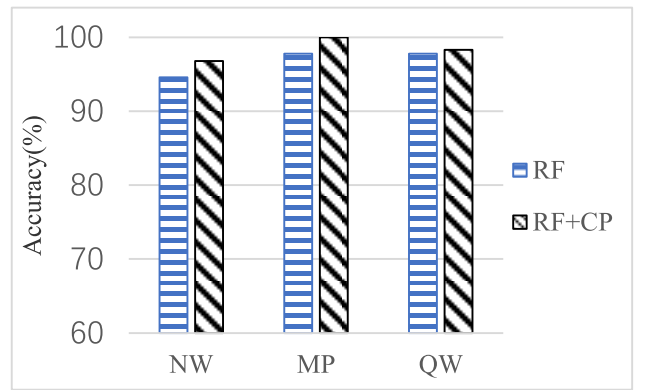


Fig. 12. Walking pattern recognition performance.

performs best in all three walking patterns, whose errors are only 1.1%, 2.9%, and 2.2%, respectively.

C. Walking Pattern Recognition Evaluation

The WPR results in Figure 12 show that RF can achieve a 96.8%, 100%, and 98.3% average recognition accuracy in NW, MP, and QW, respectively. Such recognition accuracy comes from the fact that our SD method can accurately detect each step boundary in different walking patterns, and the features extracted at each step have a good degree of discrimination. Moreover, CP also plays a big role, improving the recognition accuracy by 1.7% on average.

Tables VI, VII, and VIII show the confusion matrixes of classifiers 1, 2, and 3. We can see that the recognition of the smartphone pose is 100% correct. The worst case of misclassification occurs between NW and MP when the smartphone is in Pose2. This is because the distinction between NW and MP in Pose2 mainly depends on the features of separate three-axis accelerations. However, it is difficult to guarantee that the smartphone's three-axis direction has been unchanged due to the vibration during motion, and this results in the misclassification of NW and MP.

D. Stride Length Estimation Evaluation

In this experiment, we divided the previously collected NW and QW training data into two parts: one for training the SLE

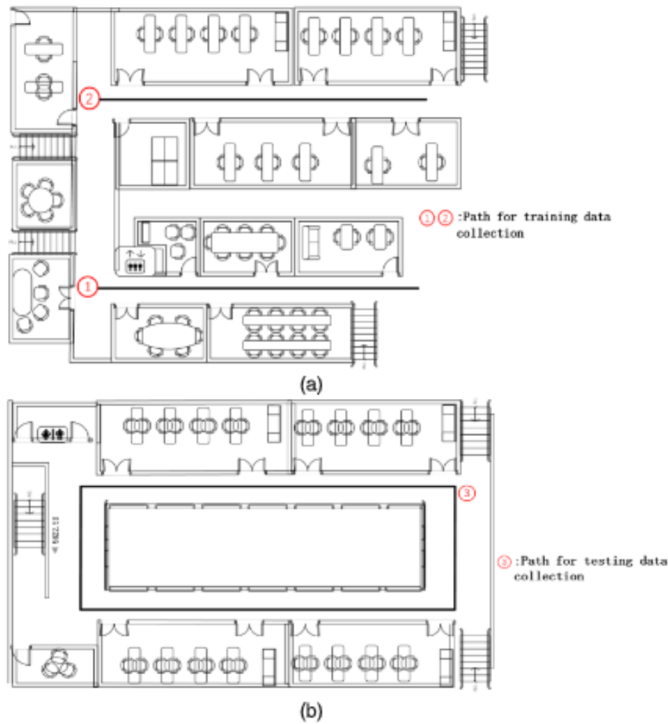


图 10.实验场地的室内布局。a) 训练数据采集发生在第一栋建筑。b) 测试数据采集发生在第六栋建筑。

表五

PS O DTW-B SD M	
Parameters	value
φ	40
τ	20
θ	0.4

我们采用相对误差来比较标清性能，定义如下：

$$SD = \frac{|N - N'|}{N} \times 100\% \quad (18)$$

其中 N 和 N' 分别是估计值和真实值。我们计算了所有测试轨迹的平均相对误差，结果如图 11 所示。

图 11 显示，我们的 SD 方法优于两种比较方法。Zhang 的方法显然不适用于 MP，因为 MP 错误率明显高于其他两种步行模式。此外，步行过程中智能手机姿势的变化使 Zhang 方法中的阈值失效，从而导致 QW 模式的误差。尽管 Gu 的方法在 NW 模式中具有可接受的检测精度，但其 SD 结果在 MP 和 QW 模式中变差。因此，Zhang 和 Gu 的方法都无法适应不同的步行模式。我们提出的方法使用 DTW 进行峰值预测，结合周期性约束和自适应阈值来适应不同的步行模式，并且

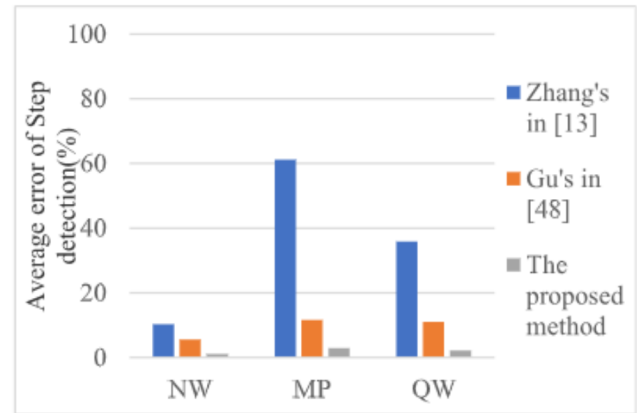


图 11.步进检测性能对比。

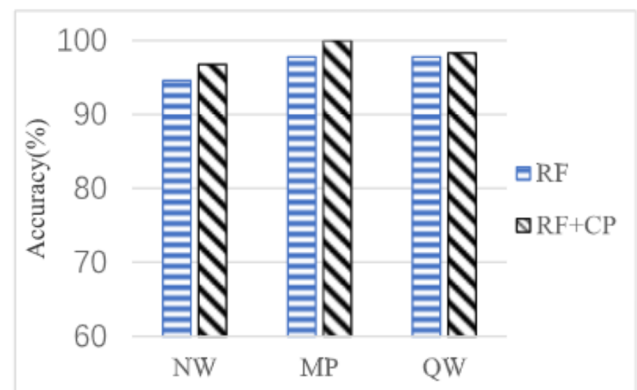


图 12.行走模式识别性能。

在所有三种步行模式中表现最佳，其误差分别仅为 1.1%、2.9% 和 2.2%。

C. 步行模式识别评估

图 12 中的 WPR 结果表明，RF 在 NW、MP 和 QW 中的平均识别准确率分别为 96.8%、100% 和 98.3%。这种识别精度来自于我们的 SD 方法能够准确检测不同行走模式下的每一步边界，并且每一步提取的特征具有良好的辨别力。此外，CP 也发挥了很大的作用，平均提高了 1.7% 的识别精度。

表 VI、VII 和 VIII 显示了分类器 1、2 和 3 的混淆矩阵。我们可以看到，智能手机姿态的识别是 100% 正确的。当智能手机处于 Pose2 时，最坏的错误分类发生在 NW 和 MP 之间。这是因为 Pose2 中 NW 和 MP 的区别主要取决于单独三轴加速度的特征。然而，由于运动过程中的振动，很难保证智能手机的三轴方向没有变化，这导致了 NW 和 MP 的错误分类。

D. 步幅估计评估

在本实验中，我们将之前收集的 NW 和 QW 训练数据分为两部分：一部分用于训练 SLE

TABLE VI
THE CONFUSION MATRIX OF CLASSIFIER 1

The actual smartphone pose	The predicted smartphone pose	
	Pose1	Pose2
Pose1	100%	0%
Pose2	0%	100%

TABLE VII
THE CONFUSION MATRIX OF CLASSIFIER 2

The actual walking pattern	The predicted walking pattern		
	NW	MP	QW
NW	97.2%	2.8%	0%
MP	1.3%	98.7%	0%
QW	0.7%	2.5%	96.8%

TABLE VIII
THE CONFUSION MATRIX OF CLASSIFIER 3

The actual walking pattern	The predicted walking pattern		
	NW	MP	QW
NW	93.3%	6.7%	0%
MP	2.9%	97.1%	0%
QW	0.9%	0.8%	98.3%

parameters and the other for testing the SLE performance. In addition, the data of different phone poses and walking patterns is processed separately and, hence, there are four combination modes in total.

Our proposed SLE is compared with Weinberg's in [32] and Shin's in [36]. All three methods have the same step parameter training process and different step parameters in each combination mode. The cumulative distribution function (CDF) of the estimation error is used to indicate the estimation accuracy. As described in Figure 13, we can see that our proposed SLE's estimation error has a probability of 90% less than 0.04 m, which is better than Shin's and Weinberg's. The reason for the higher accuracy of our proposed SLE method is that it uses more characteristics for modeling.

E. Walking Distance Estimation Evaluation

In this experiment, we also adopted the relative error of the estimated walking distance, as shown in (19), to measure the performance.

$$WDe = \frac{|D_e - D_t|}{D_t} \quad (19)$$

where D_e and D_t are the estimated and the true walking distance, respectively. The estimated walking distance is calculated by summing all the steps for each testing trajectory:

$$D_e = \sum_{i=1}^N SL_i \quad (20)$$

where N is the number of steps, and SL_i is each step's stride length. Since the calculation of the walking distance requires

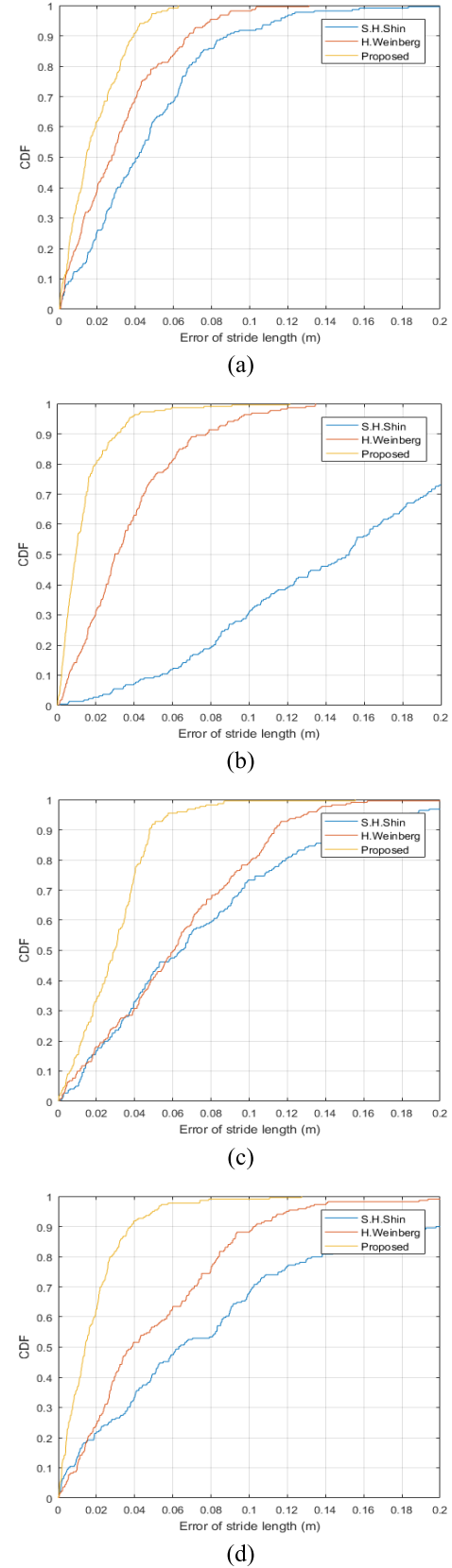


Fig. 13. The CDF of the stride length error in (a) NW in pose1, (b) QW in pose1, (c) NW in pose2, and (d) QW in pose2.

both SD and SLE, we combined Zhang's [13] and Gu's [48] SD methods with Weinberg's [32] SLE method as comparison algorithms, respectively. In addition, we also compared the

表六

TCMC1		
The actual smartphone pose	The predicted smartphone pose	
	Pose1	Pose2
Pose1	100%	0%
Pose2	0%	100%

表七

TCMC2			
The actual walking pattern	The predicted walking pattern		
	NW	MP	QW
NW	97.2%	2.8%	0%
MP	1.3%	98.7%	0%
QW	0.7%	2.5%	96.8%

表八

TCMC3			
The actual walking pattern	The predicted walking pattern		
	NW	MP	QW
NW	93.3%	6.7%	0%
MP	2.9%	97.1%	0%
QW	0.9%	0.8%	98.3%

参数和另一个用于测试系统性红斑狼疮性能。此外, 不同手机姿势和行走模式的数据是单独处理的, 因此总共有四种组合模式。我们提出的 SLE 与[32]中的 Weinberg 和[36]中的 Shin 进行了比较。这三种方法在每种组合模式下具有相同的步进参数训练过程和不同的步进参数。估计误差的累积分布函数 (CDF) 用于指示估计精度。如图 13 所示, 我们可以看到我们提出的 SLE 估计误差的概率小于 0.04 m 的概率为 90%, 优于 Shin 和 Weinberg 的。我们提出的 SLE 方法精度较高的原因是它使用了更多的特征进行建模。

E. 步行距离估计评估

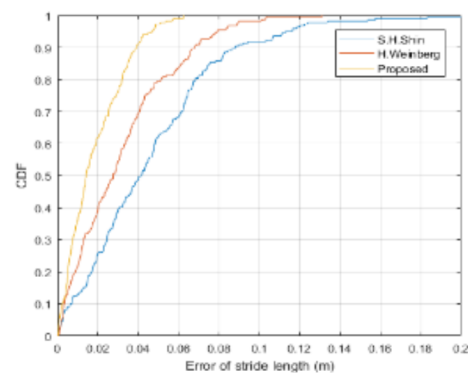
在本实验中, 我们还采用了估计步行距离的相对误差, 如 (19) 所示, 来衡量性能。

$$\text{在 } D = \frac{|D - D|}{D} \quad (19)$$

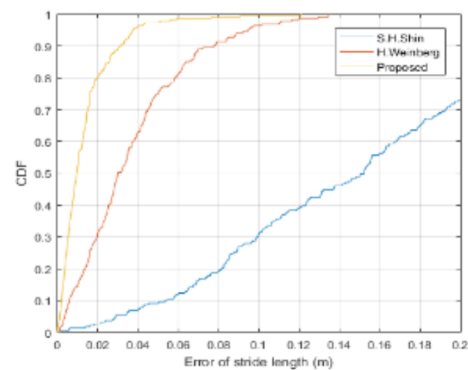
其中 D 和 D 分别是估计的步行距离和真实步行距离。估计的步行距离是通过将每个测试轨迹的所有步数相加来计算的:

$$D = \sum_{i=1}^N \text{小号 } L \quad (20)$$

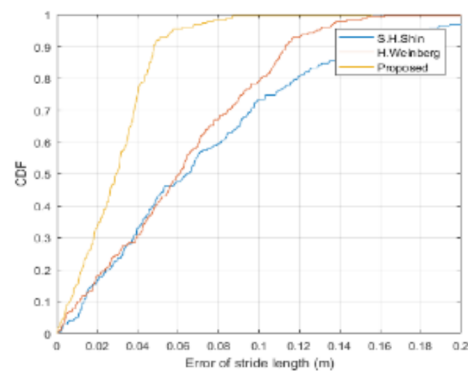
其中 N 是步数, SL 是每步的步幅。由于步行距离的计算需要



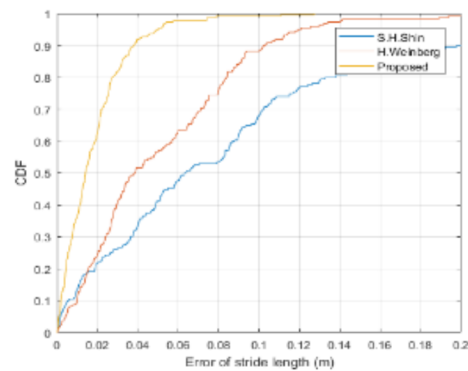
(a)



(b)



(c)



(d)

图 13. (a) 姿势 1 中的 NW、(b) 姿势 1 中的 QW、(c) 姿势 2 中的 NW 和 (d) 姿势 2 中的 QW 的步幅误差的 CDF。

对于 SD 和 SLE, 我们分别将 Zhang 的[13]和 Gu 的[48]SD 方法与 Weinberg 的[32]SLE 方法相结合作为比较算法。此外, 我们

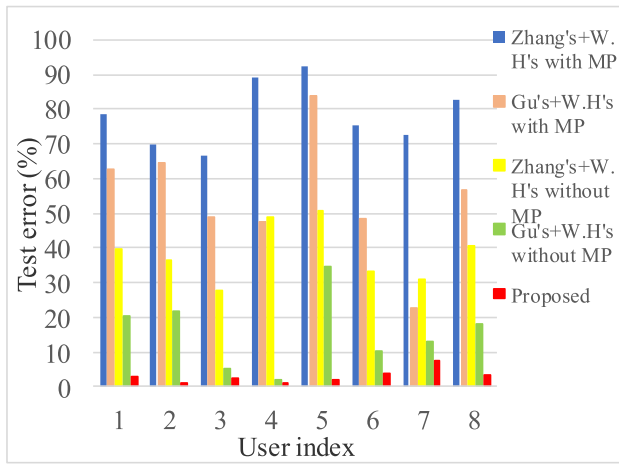


Fig. 14. Performance comparison with commonly used methods.

performance of algorithms with and without the MP mode. The final comparison results are shown in Figure 14.

From Figure 14, it can be seen that our method outperforms the commonly used walking distance estimation methods. For all the users, our method can achieve the best estimation accuracy, and its average relative error is 3.0%. Since the first two comparison algorithms cannot recognize the MP mode, their walking distance estimation errors are very large. This indicates that if the pedestrian adopts an MP pattern during the walking process, it is very important to identify MP for the walking distance estimation. Although the latter two comparison algorithms do not consider the influence of MP on the walking distance estimation, their walking distance estimation errors are still larger than that of our proposed method. The reason why our proposed method works well lies in three aspects. First, our method's SD results are more accurate, as shown in Figure 11. Second, our SLE model is a fusion of two algorithms, which is better than the comparison algorithm, as shown in Figure 13. Third, we distinguished walking patterns and trained different SLE parameters for NW and QW.

V. CONCLUSION

In this paper, we propose a robust SD and SLE method for PDR based on recognizing three walking patterns using the smartphone's built-in inertial sensors. Our proposed method includes a novel SD, an RF-based WPR, and a walking pattern-aware SLE. In particular, the proposed SD fuses DTW-based peak prediction and zero-crossing detection, and thus it can accurately identify each step's starting and ending points. Then, on the basis of the extracted features from each step, three RF-based classifications with CP are used to recognize two phone poses and three pedestrian walking patterns. According to the recognition results of the walking patterns, the proposed SLE method can adaptively select the appropriate SLE model, avoiding the large SLE errors caused by unknown walking patterns. By comparing our method with the existing commonly used methods, we show our method's superiority.

In the future, we will consider more walking patterns and phone poses, such as walking upstairs, walking downstairs, and swinging the phone in hand, to improve its application range. In addition, we will also invite more volunteers to participate in experiments to further improve our method's generalization and practicability.

ACKNOWLEDGMENT

The authors would like to thank the editors and the anonymous reviewers for their constructive comments and suggestions, which helped to improve the quality of this paper.

REFERENCES

- [1] R. S. Nerem and K. M. Larson, "Global positioning system, theory and practice, 5th edition," *Eos, Trans. Amer. Geophys. Union*, vol. 82, no. 33, p. 365, 2013.
- [2] Y. Sun, M. Liu, and M. Q.-H. Meng, "WiFi signal strength-based robot indoor localization," in *Proc. IEEE Int. Conf. Inf. Automat. (ICIA)*, Jul. 2014, pp. 250–256.
- [3] A. Athalye, V. Savic, M. Bolic, and P. M. Djuric, "A radio frequency identification system for accurate indoor localization," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Prague, Czech Republic, May 2011, pp. 1777–1780.
- [4] F. Despaux, A. van den Bossche, K. Jaffrès-Runser, and T. Val, "N-TWR: An accurate time-of-flight-based N-ary ranging protocol for ultra-wide band," *Ad Hoc Netw.*, vol. 79, pp. 1–19, Oct. 2018.
- [5] K. Lee, Y. Nam, and S. D. Min, "An indoor localization solution using Bluetooth RSSI and multiple sensors on a smartphone," *Multimedia Tools Appl.*, vol. 77, no. 10, pp. 12635–12654, May 2018.
- [6] B. Zhou, Q. Li, Q. Mao, W. Tu, and X. Zhang, "Activity sequence-based indoor pedestrian localization using smartphones," *IEEE Trans. Human-Mach. Syst.*, vol. 45, no. 5, pp. 562–574, Oct. 2015.
- [7] W. Kang and Y. Han, "SmartPDR: Smartphone-based pedestrian dead reckoning for indoor localization," *IEEE Sensors J.*, vol. 15, no. 5, pp. 2906–2916, May 2015.
- [8] L. Ciabattini *et al.*, "Real time indoor localization integrating a model based pedestrian dead reckoning on smartphone and BLE beacons," *J. Ambient Intell. Hum. Comput.*, vol. 10, no. 1, pp. 1–12, Jan. 2019.
- [9] A. Abadleh, E. Al-Hawari, E. Alkafaween, and H. Al-Sawalqah, "Step detection algorithm for accurate distance estimation using dynamic step length," in *Proc. 18th IEEE Int. Conf. Mobile Data Manage. (MDM)*, May 2017, pp. 324–327.
- [10] P. Chen, Y. Kuang, and X. Chen, "A UWB/improved PDR integration algorithm applied to dynamic indoor positioning for pedestrians," *Sensors*, vol. 17, no. 9, pp. 2065–2084, 2017.
- [11] F. Weimann, G. Abwerzger, and B. Hofmann-Wellenhof, "A pedestrian navigation system for urban and indoor environments," in *Proc. Int. Tech. Meeting Satell. Division Inst. Navigat.*, 2007, pp. 1380–1389.
- [12] M.-S. Pan and H.-W. Lin, "A step counting algorithm for smartphone users: Design and implementation," *IEEE Sensors J.*, vol. 15, no. 4, pp. 2296–2305, Apr. 2015.
- [13] M. Zhang, Y. Wen, J. Chen, X. Yang, R. Gao, and H. Zhao, "Pedestrian dead-reckoning indoor localization based on OS-ELM," *IEEE Access*, vol. 6, pp. 6116–6129, 2018.
- [14] A. Brajdic and R. Harle, "Walk detection and step counting on unconstrained smartphones," in *Proc. ACM Int. Joint Conf. Pervas. Ubiquitous Comput. (UbiComp)*, 2013, pp. 225–234.
- [15] S. Del Din, A. Godfrey, and L. Rochester, "Validation of an accelerometer to quantify a comprehensive battery of gait characteristics in healthy older adults and Parkinson's disease: Toward clinical and at home use," *IEEE J. Biomed. Health Informat.*, vol. 20, no. 3, pp. 838–847, May 2016.
- [16] M. H. Pham *et al.*, "Validation of a step detection algorithm during straight walking and turning in patients with Parkinson's disease and older adults using an inertial measurement unit at the lower back," *Frontiers Neurol.*, vol. 8, pp. 457–465, Sep. 2017.
- [17] J.-H. Wang, J.-J. Ding, Y. Chen, and H.-H. Chen, "Real time accelerometer-based gait recognition using adaptive windowed wavelet transforms," in *Proc. IEEE Asia Pacific Conf. Circuits Syst.*, Dec. 2012, pp. 591–594.
- [18] Y. Cho, H. Cho, and C.-M. Kyung, "Design and implementation of practical step detection algorithm for wrist-worn devices," *IEEE Sensors J.*, vol. 16, no. 21, pp. 7720–7730, Nov. 2016.

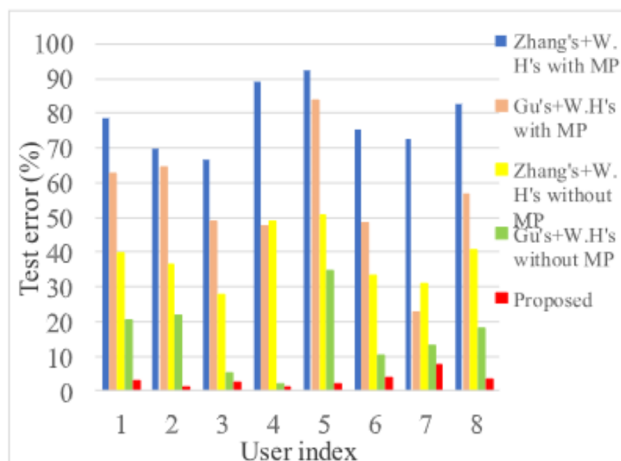


图 14.与常用方法的性能比较。

带和不带 MP 模式的算法性能。

最终的比较结果如图 14 所示。

从图 14 可以看出，我们的方法优于常用的步行距离估计方法。对于所有用户，我们的方法都能达到最佳的估计精度，其平均相对误差为 3.0%。由于前两种比较算法无法识别 MP 模式，因此它们的步行距离估计误差非常大。这表明，如果行人在行走过程中采用 MP 模式，那么识别 MP 对于步行距离估计非常重要。虽然后两种比较算法没有考虑 MP 对步行距离估计的影响，但它们的步行距离估计误差仍然大于我们所提出的方法。我们所提出的方法之所以效果很好，原因在于三个方面。首先，我们方法的 SD 结果更准确，如图 11 所示。其次，我们的 SLE 模型是两种算法的融合，优于比较算法，如图 13 所示。第三，区分行走模式，训练 NW 和 QW 不同的 SLE 参数。

五、C

在本文中，我们提出了一种基于使用智能手机内置惯性传感器识别三种步行模式的鲁棒 SD 和 SLEPDR 方法。我们提出的方法包括一种新型 SD、一种基于 RF 的 WPR 和一种步行模式感知 SLE。特别是，所提出的 SD 融合了基于 DTW 的峰值预测和过零检测，因此可以准确识别每个步骤的起点和终点。然后，在从每个步骤中提取的特征的基础上，使用具有 CP 的三种基于 RF 的分类来识别两个手机姿态和三种行人步行模式。根据步行模式的识别结果，所提出的 SLE 方法可以自适应地选择合适的 SLE 模型，避免了未知步行模式引起的较大 SLE 误差。通过将我们的方法与现有的常用方法进行比较，我们展示了该方法的优越性。

未来，我们将考虑更多的行走模式和手机姿势，如上楼、下楼、手中摆动手机等，以提高其应用范围。此外，我们还将邀请更多的志愿者参与实验，以进一步提高我们方法的通用性和实用性。

A

作者在此感谢编辑和匿名审稿人的建设性意见和建议，有助于提高本文的质量。

R

- [1] R. S. Nerem 和 K. M. Larson, “全球定位系统, 理论与实践, 第 5 版”, Eos, Trans. Amer. Geophys. Union, 第 82 卷, 第 33 期, 第 365 页, 2013 年。[2] Y. Sun, M. Liu 和 M. Q.-H. Meng, “基于 WiFi 信号强度的机器人室内定位”, 载于 Proc. IEEE Int. Conf. Inf. Automat. (ICIA), 2014 年 7 月, 第 250-256 页。
- [3] A. Athalye, V. Savic, M. Bolic 和 P. M. Djuric, “用于准确室内定位的射频识别系统”, Proc. IEEE Int. Conf. Acoust., 语音信号过程. (ICASSP), 捷克共和国布拉格, 2011 年 5 月, 第 1777-1780 页。
- [4] F. Despaux, A. van den Bossche, K. Jaffrès-Runser 和 T. Val, “N-TWR: 一种用于超宽带的基于飞行时间的精确 N-ary ranging protocol”, Ad Hoc Netw., 第 79 卷, 第 1-19 页, 2018 年 10 月。[5] K. Lee, Y. Nam 和 S. D. Min, “在智能手机上使用蓝牙 RSSI 和多个传感器的室内定位解决方案”, Multimedia Tools Appl., 第 77 卷, 第 10 期, 第 12635-12654 页, 2018 年 5 月。[6] B. 周, Q. Li, Q. 毛, W. Tu 和 X. Zhang, “使用智能手机的基于活动序列的室内行人定位”, IEEE Trans. 人类马赫. Syst., 第 45 卷, 第 5 期, 第 562-574 页, 2015 年 10 月。
- [7] W. Kang 和 Y. Han, “SmartPDR: 用于室内定位的基于智能手机的行人航位推算”, IEEE Sensors J., 第 15 卷, 第 5 期, 第 2906-2916 页, 2015 年 5 月。
- [8] L. Ciabattini 等人, “在智能手机和 BLE 信标上集成基于模型的行人航位推算的实时室内定位”, J. 环境智能. Hum. Comput., 第 10 卷, 第 1 期, 第 1-12 页, 2019 年 1 月。
- [9] A. Abadleh, E. Al-Hawari, E. Alkafaween 和 H. Al-Sawalqah, “使用动态步进进行精确距离估计的步进检测算法”, 第 18 届 IEEE 国际会议移动数据管理论文集. (MDM), 2017 年 5 月, 第 324-327 页。
- [10] P. Chen, Y. Kuang 和 X. Chen, “UWB/改进的 PDR 集成应用于行人动态室内定位的算法”, Sensors, 第 17 卷, 第 9 期, 第 2065-2084 页, 2017 年。
- [11] F. Weimann, G. Abwerzger 和 B. Hofmann-Wellenhof, “用于城市和室内环境的行人导航系统”, 载于 Proc. Int. 技术会议萨特尔. Division Inst. Navigat., 2007 年, 第 1380-1389 页。
- [12] M.-S. 潘和 H.-W. Lin, “智能手机的计步算法”, 设计 and 实现, IEEE Sensors J., 第 15 卷, 第 4 期, 第 2296-2305 页, 2015 年 4 月。
- [13] M. Zhang, Y. 温, J. Chen, X. Yang, R. Gao, and H. Zhao, “Pedestrian 基于 OS-ELM 的航位推算室内定位”, IEEE Access, 第 6 卷, 第 6116-6129 页, 2018 年。
- [14] A. Brajdic 和 R. Harle, “步行检测和步数计算”, 紧张的手机, 载于 Proc. ACM Int. Joint Conf. Pervas. 无处不在的计算. UbiComp), 2013 年, 第 225-234 页。
- [15] S. Del Din, A. Godfrey 和 L. Rochester, “加速度的验证-量化健康老年人和帕金森病的一系列综合步态特征: 走向临床和家庭使用”, IEEE J. Biomed. 健康信息, 第 20 卷, 第 3 期, 第 838-847 页, 2016 年 5 月。
- [16] MH Pham 等人, “使用下背部惯性测量装置在帕金森病患者和老年人中直行和转身”, Frontiers Neurol., 第 8 卷, 第 457-465 页, 2017 年 9 月。
- [17] J.-H. 王 J.-J. 丁 Y. Chen 和 H.-H. 陈, “使用自适应窗口小波变换的基于加速度计的步态识别”, 载于 IEEE Asia Pacific Conf. Circuits Syst., 2012 年 12 月, 第 591-594 页。
- [18] Y. Cho, H. Cho 和 C.-M. Kyung, “设计和实施”, 腕戴式设备的实用步进检测算法, IEEE Sensors J., 第 16 卷, 第 21 期, 第 7720-7730 页, 2016 年 11 月。

- [19] S. Kammoun, J.-B. Pothin, and J.-C. Cousin, "An efficient fuzzy logic step detection algorithm for unconstrained smartphones," in *Proc. IEEE 26th Annu. Int. Symp. Pers., Indoor, Mobile Radio Commun. (PIMRC)*, Aug. 2015, pp. 2110–2114.
- [20] X. Kang, B. Huang, and G. Qi, "A novel walking detection and step counting algorithm using unconstrained smartphones," *Sensors*, vol. 18, no. 1, pp. 297–312, 2018.
- [21] H. Li and L. Yang, "Accurate and fast dynamic time warping," in *Proc. Int. Conf. Adv. Data Mining Appl.*, 2013, pp. 133–144.
- [22] M. Susi, V. Renaudin, and G. Lachapelle, "Detection of quasi-static instants from handheld MEMS devices," in *Proc. Int. Conf. Indoor Positioning Indoor Navigat.*, Sep. 2011, pp. 1–9.
- [23] X. Niu, M. Li, X. Cui, J. Liu, S. Liu, and K. R. Chowdhury, "WTrack: HMM-based walk pattern recognition and indoor pedestrian tracking using phone inertial sensors," *Pers. Ubiquitous Comput.*, vol. 18, no. 8, pp. 1901–1915, Dec. 2014.
- [24] B. Wang, X. Liu, B. Yu, R. Jia, and X. Gan, "Pedestrian dead reckoning based on motion mode recognition using a smartphone," *Sensors*, vol. 18, no. 6, pp. 1811–1835, 2018.
- [25] C. Catal, S. Tufekci, E. Pirmitt, and G. Kocabag, "On the use of ensemble of classifiers for accelerometer-based activity recognition," *Appl. Soft Comput.*, vol. 37, pp. 1018–1022, Dec. 2015.
- [26] S. Miyamoto and H. Ogawa, "Human activity recognition system including smartphone position," *Procedia Technol.*, vol. 18, pp. 42–46, Jan. 2014.
- [27] F. Gu, A. Kealy, K. Khoshelham, and J. Shang, "User-independent motion state recognition using smartphone sensors," *Sensors*, vol. 15, no. 12, pp. 30636–30652, 2015.
- [28] M. M. Hassan, M. Z. Uddin, A. Mohamed, and A. Almogren, "A robust human activity recognition system using smartphone sensors and deep learning," *Future Gener. Comput. Syst.*, vol. 81, pp. 307–313, Apr. 2018.
- [29] F. Gu, K. Khoshelham, C. Yu, and J. Shang, "Accurate step length estimation for pedestrian dead reckoning localization using stacked autoencoders," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 8, pp. 2705–2713, Aug. 2019.
- [30] F. Gu, K. Khoshelham, S. Valaee, J. Shang, and R. Zhang, "Locomotion activity recognition using stacked denoising autoencoders," *IEEE Internet Things J.*, vol. 5, no. 3, pp. 2085–2093, Jun. 2018.
- [31] B. Zhou, J. Yang, and Q. Li, "Smartphone-based activity recognition for indoor localization using a convolutional neural network," *Sensors*, vol. 19, no. 3, pp. 621–635, 2019.
- [32] H. Weinberg, "Using the ADXL202 in pedometer and personal navigation applications," Analog Devices, Norwood, MA, USA, Appl. Note AN-602, 2002, pp. 1–6.
- [33] J. C. Alvarez, D. Alvarez, A. López, and R. C. González, "Pedestrian navigation based on a waist-worn inertial sensor," *Sensors*, vol. 12, no. 8, pp. 10536–10549, 2012.
- [34] W. Zhu, B. Anderson, S. Zhu, and Y. Wang, "A computer vision-based system for stride length estimation using a mobile phone camera," in *Proc. 18th Int. ACM SIGACCESS Conf. Comput. Accessibility*, Oct. 2016, pp. 121–130.
- [35] E. Martin, "Novel method for stride length estimation with body area network accelerometers," in *Proc. IEEE Topical Conf. Biomed. Wireless Technol., Netw., Sens. Syst.*, Jan. 2011, pp. 79–82.
- [36] S. H. Shin, C. G. Park, J. W. Kim, H. S. Hong, and J. M. Lee, "Adaptive step length estimation algorithm using low-cost MEMS inertial sensors," in *Proc. IEEE Sensors Appl. Symp.*, Feb. 2007, pp. 1–5.
- [37] A. Peruzzi, U. D. Croce, and A. Cereatti, "Estimation of stride length in level walking using an inertial measurement unit attached to the foot: A validation of the zero velocity assumption during stance," *J. Biomech.*, vol. 44, no. 10, pp. 1991–1994, Jul. 2011.
- [38] J. Hannink, T. Kautz, C. F. Pasluosta, K.-G. Gasmann, J. Klucken, and B. M. Eskofier, "Sensor-based gait parameter extraction with deep convolutional neural networks," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 1, pp. 85–93, Jan. 2017.
- [39] J. Hannink *et al.*, "Stride length estimation with deep learning," 2016, *arXiv:1609.03321*. <https://arxiv.org/abs/1609.03321>
- [40] J. Park, "Synthesis of natural arm swing motion in human bipedal walking," *J. Biomech.*, vol. 41, no. 7, pp. 1417–1426, Jan. 2008.
- [41] J.-S. Wang, C.-W. Lin, Y.-T.-C. Yang, and Y.-J. Ho, "Walking pattern classification and walking distance estimation algorithms using gait phase information," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 10, pp. 2884–2892, Oct. 2012.
- [42] P. Gupta and T. Dallas, "Feature selection and activity recognition system using a single triaxial accelerometer," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 6, pp. 1780–1786, Jun. 2014.
- [43] M. del Rosario, S. Redmond, and N. Lovell, "Tracking the evolution of smartphone sensing for monitoring human movement," *Sensors*, vol. 15, no. 8, pp. 18901–18933, 2015.
- [44] Y.-P. Chen, J.-Y. Yang, S.-N. Liou, G.-Y. Lee, and J.-S. Wang, "Online classifier construction algorithm for human activity detection using a tri-axial accelerometer," *Appl. Math. Comput.*, vol. 205, no. 2, pp. 849–860, Nov. 2008.
- [45] S. González, J. Sedano, J. R. Villar, E. Corchado, Á. Herrero, and B. Baruaque, "Features and models for human activity recognition," *Neurocomputing*, vol. 167, pp. 52–60, Nov. 2015.
- [46] I. Kononenko, "Estimating attributes: Analysis and extensions of RELIEF," in *Proc. Eur. Conf. Mach. Learn. Learn.*, vol. 94, 1994, pp. 171–182.
- [47] Y. Wang and F. Makedon, "Application of relief-F feature filtering algorithm to selecting informative genes for cancer classification using microarray data," in *Proc. IEEE Comput. Syst. Bioinf. Conf. (CSB)*, Aug. 2004, pp. 497–498.
- [48] F. Gu, K. Khoshelham, J. Shang, F. Yu, and Z. Wei, "Robust and accurate smartphone-based step counting for indoor localization," *IEEE Sensors J.*, vol. 17, no. 11, pp. 3453–3460, Jun. 2017.



Yingbiao Yao received the Ph.D. degree in information and communication engineering from Zhejiang University, Hangzhou, China, in 2006.

From 2008 to 2016, he was an Associate Professor with the School of Communication Engineering, Hangzhou Dianzi University, China. In 2011, he was a Visiting Scholar with RPI University, NY, USA. Since 2017, he has been a Professor with the School of communication engineering, Hangzhou Dianzi University. He is the author of three books, more than 50 articles, and more than 30 inventions. His research interests include storage system design, wireless sensor network, indoor localization, and media signal processing.

Dr. Yao was a recipient of the third level of Zhejiang Provincial "151" Talent for Excellence in 2015, the first class prize of Zhejiang Provincial Scientific and Technological Progress Award in 2016, and Zhejiang Provincial young and middle-aged academic leaders in 2017.

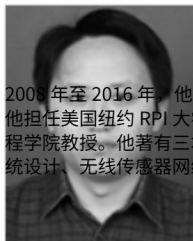


Lei Pan received the B.E. degree in communication engineering from Hangzhou Dianzi University, China, in 2017, where he is currently pursuing the M.E. degree. His current research interest is indoor localization technology.



Wei Feng received the B.E. degree in electronics and information engineering from Hubei Engineering University, China, in 2005, and the M.E. and Ph.D. degrees in communication and information systems from the South China University of Technology, China, in 2009 and 2014, respectively. From 2005 to 2006, she was an FAE with LITE-ON Technology Cooperation, Guangzhou, China. She is currently a Lecturer with Hangzhou Dianzi University. Her research interests emphasize on energy efficiency, indoor localization technology, and physical layer security in future wireless communications.

- [19] S. Kammoun, J.-B. Pothin 和 J.-C. Cousin, “一种有效的模糊逻辑无约束智能手机的步进检测算法”, 载于 IEEE 第 26 届年鉴论文集. Int. Symp. Pers., 室内, 移动无线电报. PIMRC), 2015 年 8 月, 第 2110–2114 页。
- [20] X. Kang, B. Huang 和 G. Qi, “一种新颖的步行检测和步进使用不受约束的智能手机进行计数算法”, Sensors, 第 18 卷, 第 1 期, 第 297–312 页, 2018 年。
- [21] H. Li 和 L. Yang, “准确和快速的动态时间扭曲”, 载于 Proc. 国际会议. 数据挖掘应用, 2013 年, 第 133–144 页。
- [22] M. Susi, V. Renaudin 和 G. Lachapelle, “准静态检测手持式 MEMS 设备的瞬间”, 载于 Proc. Int. Conf. Indoor Positioning Indoor Navigat., 2011 年 9 月, 第 1–9 页。
- [23] X. Niu, M. Li, X. Cui, J. Liu, S. Liu 和 K. R. Chowdhury, “WTrack: 使用手机惯性传感器的基于 HMM 的步行模式识别和室内行人跟踪”, Pers. Ubiquitous Comput., 第 18 卷, 第 8 期, 第 1901–1915 页, 2014 年 12 月。
- [24] B. Wang, X. Liu, B. Yu, R. Jia 和 X. Gan, “Pedestrian dead reckoning 基于使用智能手机的运动模式识别”, Sensors, 第 18 卷, 第 6 期, 第 1811–1835 页, 2018 年。
- [25] C. Catal, S. Tufekci, E. Pirmitt 和 G. Kocabag, “关于合奏的使用基于加速度计的活动识别的分类器”, Appl. Soft Comput., 第 37 卷, 第 1018–1022 页, 2015 年 12 月。
- [26] S. Miyamoto 和 H. Ogawa, “人类活动识别系统包括智能手机位置”, Procedia Technol., 第 18 卷, 第 42–46 页, 2014 年 1 月。
- [27] F. Gu, A. Kealy, K. Khoshelham 和 J. Shang, “与用户无关使用智能手机传感器进行运动状态识别”, 传感器, 第 15 卷, 第 12 期, 第 30636–30652 页, 2015 年。
- [28] M. M. Hassan, M. Z. Uddin, A. Mohamed 和 A. Almogren, “一个强大的使用智能手机传感器和深度学习的人类活动识别系统”, 未来一代. 计算. 系统, 第 81 卷, 第 307–313 页, 2018 年 4 月。
- [29] F. Gu, K. Khoshelham, C. Yu 和 J. Shang, “准确的步长估计使用堆叠自动编码器进行行人航位推算定位的配合”, IEEE Trans. Instrum. Meas., 第 68 卷, 第 8 期, 第 2705–2713 页, 2019 年 8 月。
- [30] F. Gu, K. Khoshelham, S. Valaee, J. Shang 和 R. Zhang, “Locomo-使用堆叠去噪自动编码器的活动识别”, IEEE Internet Things J., 第 5 卷, 第 3 期, 第 2085–2093 页, 2018 年 6 月。
- [31] B. 周, J. Yang 和 Q. Li, “基于智能手机的活动识别使用卷积神经网络进行室内定位”, Sensors, 第 19 卷, 第 3 期, 第 621–635 页, 2019 年。
- [32] H. Weinberg, “在计步器和个人导航应用中使用 ADXL202”, ADI 公司, 美国马萨诸塞州诺伍德, 应用. 注释 AN-602. 2002 年, 第 1–6 页。
- [33] J. C. Alvarez, D. Alvarez, A. López 和 R. C. González, “行人基于腰部惯性传感器的导航”, 传感器, 第 12 卷, 第 8 期, 第 10536–10549 页, 2012 年。
- [34] W. Zhu, B. Anderson, S. Zhu 和 Y. Wang, “一种基于计算机视觉的系统, 用于使用手机摄像头进行步幅估计”, 第 18 届国际会议记录. 无障碍, 2016 年 10 月, 第 121–130 页。
- [35] E. Martin, “用身体面积估计步幅的新方法网络加速度计”, 载于 IEEE Topical Conf. Biomed. Wireless Technol., Netw., Sens. Syst., 2011 年 1 月, 第 79–82 页。
- [36] S. H. Shin, C. G. Park, J. W. Kim, H. S. Hong, and J. M. Lee, “Adaptive steplength estimation algorithm using low-cost MEMS inertial sensors”, IEEE Sensors Appl. Symp., 2007 年 2 月, 第 1–5 页。
- [37] A. Peruzzi, U. D. Croce 和 A. Cereatti, “步幅估计使用连接到脚部的惯性测量单元进行水平行走: 站立期间零速度假设的验证”, J. Biomech., 第 44 卷, 第 10 期, 第 1991–1994 页, 2011 年 7 月。
- [38] J. 汉宁克, T. 考茨, CF 帕斯洛斯塔, K.-G. 加斯曼, J. 克鲁肯, 和 B. M. Eskofier, “使用深度卷积神经网络提取基于传感器的步态参数提取”, IEEE J. Biomed. 健康信息, 第 21 卷, 第 1 期, 第 85–93 页, 2017 年 1 月。
- [39] J. Hannink 等人, “深度学习的步幅估计”, 2016 年, arXiv: 1609.03321. <https://arxiv.org/abs/1609.03321>
- [40] J. Park, “人类双足行走中自然手臂摆动运动的合成”, J. Biomech., 第 41 卷, 第 7 期, 第 1417–1426 页, 2008 年 1 月。[41] J.-S. 王, C.-W. 林, Y.-T.-C. 杨和 Y.-J. 何, “步行拍打-使用步态阶段信息的燕鸥分类和步行距离估计算法”, IEEE Trans. Biomed. Eng., 第 59 卷, 第 10 期, 第 2884–2892 页, 2012 年 10 月。
- [42] P. Gupta 和 T. Dallas, “特征选择和活动识别使用单个三轴加速度计的系统”, IEEE Trans. Biomed. Eng., 第 61 卷, 第 6 期, 第 1780–1786 页, 2014 年 6 月。
- [43] M. del Rosario, S. Redmond 和 N. Lovell, “追踪用于监控人体运动的智能手机传感”, Sensors, 第 15 卷, 第 8 期, 第 18901–18933 页, 2015 年。
- [44] Y.-P. 陈, J.-Y. 杨, S.-N. 刘, G.-Y. 李, J.-S. 王, “在线使用三轴加速度计进行人体活动检测的分类器构建算法”, Appl. Math. Comput., 第 205 卷, 第 2 期, 第 849–860 页, 2008 年 11 月。
- [45] S. González, J. Sedano, J. R. Villar, E. Corchado, Á. Herrero 和 B. Baruaque, “人类活动识别的特征和模型”, 神经计算, 第 167 卷, 第 52–60 页, 2015 年 11 月。
- [46] I. Kononenko, “估计属性: 分析和扩展 RELIEF”, 载于 Proc. Eur. Conf. Mach. Learn. Learn., 第 94 卷, 1994 年, 第 171–182 页。
- [47] Y. Wang 和 F. Makedon, “浮雕 F 特征滤波的应用使用微阵列数据选择癌症分类信息基因的算法”, 载于 Proc. IEEE Comput. 系统生物 inf. 会议 (CSB), 2004 年 8 月, 第 497–498 页。
- [48] F. Gu, K. Khoshelham, J. Shang, F. Yu 和 Z. Wei, “稳健而准确基于智能手机的室内定位计步”, IEEE Sensors J., 第 17 卷, 第 11 期, 第 3453–3460 页, 2017 年 6 月。



姚颖彪于 2006 年获得浙江大学信息与通信工程博士学位。

2008 年至 2016 年, 他担任中国杭州电子科技大学通信工程学院副教授。2011 年, 他担任美国纽约 RPI 大学访问学者。自 2017 年起, 他担任杭州电子科技大学通信工程学院教授。他著有 3 本书, 50 多篇文章, 30 多项发明。他的研究兴趣包括存储系统设计、无线传感器网络、室内定位和媒体信号处理。

姚博士是浙江省“151”三级获得者
2015 年优秀人才, 2016 年浙江省科技进步一等奖, 2017 年浙江省中青年学术带头人。



潘磊于 2017 年获得中国杭州电子科技大学通信工程学士学位, 目前正在该校攻读硕士学位。他目前的研究兴趣是室内定位技术。

Wei Feng 于 2005 年获得中国湖北工程大学电子与信息工程学士学位, 并分别于 2009 年和 2014 年获得中国华南理工大学通信与信息系统硕士和博士学位。2005 年至 2016 年, 她是中国广州光宝技术合作部的 FAE。她目前是杭州电子科技大学的讲师。她的研究兴趣侧重于未来无线通信中的能源效率、室内定位技术和物理层安全。





Xiaorong Xu (Member, IEEE) received the Ph.D. degree in signal and information processing from the Nanjing University of Posts and Telecommunications, Nanjing, China, in 2010. From 2011 to 2013, he was working as a Postdoctoral Researcher with the Institute of Information and Communication Engineering, Zhejiang University, Hangzhou, China. From 2013 to 2014, he serves as a Research Scholar with the Electrical and Computer Engineering Department, Stevens Institute of Technology, Hoboken, NJ, USA. He is currently working as an Associate Professor and he is an Excellent Backbone Teacher with the School of Communication Engineering, Hangzhou Dianzi University, Hangzhou, China. His research interests emphasize on wireless and mobile communications, signal processing, and storage system design.

Xuesong Liang, photograph and biography not available at the time of publication.



Xin Xu received the Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2001. From 2002 to 2011, he was an Associate Professor with the National University of Defense Technology. Since 2012, he has been a Professor with the National University of Defense Technology and Hangzhou Dianzi University. He is the author of 2 books and more than 30 articles. His research interests include solid-state storage system design, instant communication, indoor localization technology, and information safety.

徐晓荣 (IEEE 会员) 于 2010 年获得南京邮电大学信号与信息处理博士学位。2011 年至 2013 年, 他在中国杭州浙江大学信息与通信工程研究所担任博士后研究员。2013 年至 2014 年, 他在美国新泽西州霍博肯市史蒂文斯理工学院电气与计算机工程系担任研究学者。他目前是中国杭州电子科技大学通信工程学院的副教授和优秀骨干教师。他的研究兴趣侧重于无线和移动通信、信号处理和存储系统设计。



梁雪松, 照片和传记在出版时不可用。

徐昕于 2001 年获得中国长沙国防科技大学信息与通信工程博士学位。2002 年至 2011 年, 任国防科技大学副教授。2012 年至今, 任国防科技大学、杭州电子科技大学教授。著书 2 部, 文章 30 余篇。研究方向包括固态存储系统设计、即时通信、室内定位技术、信息安全等。

